AWS INTERNSHIP PROJECT REPORT

VPC Peering & Private Network Connectivity

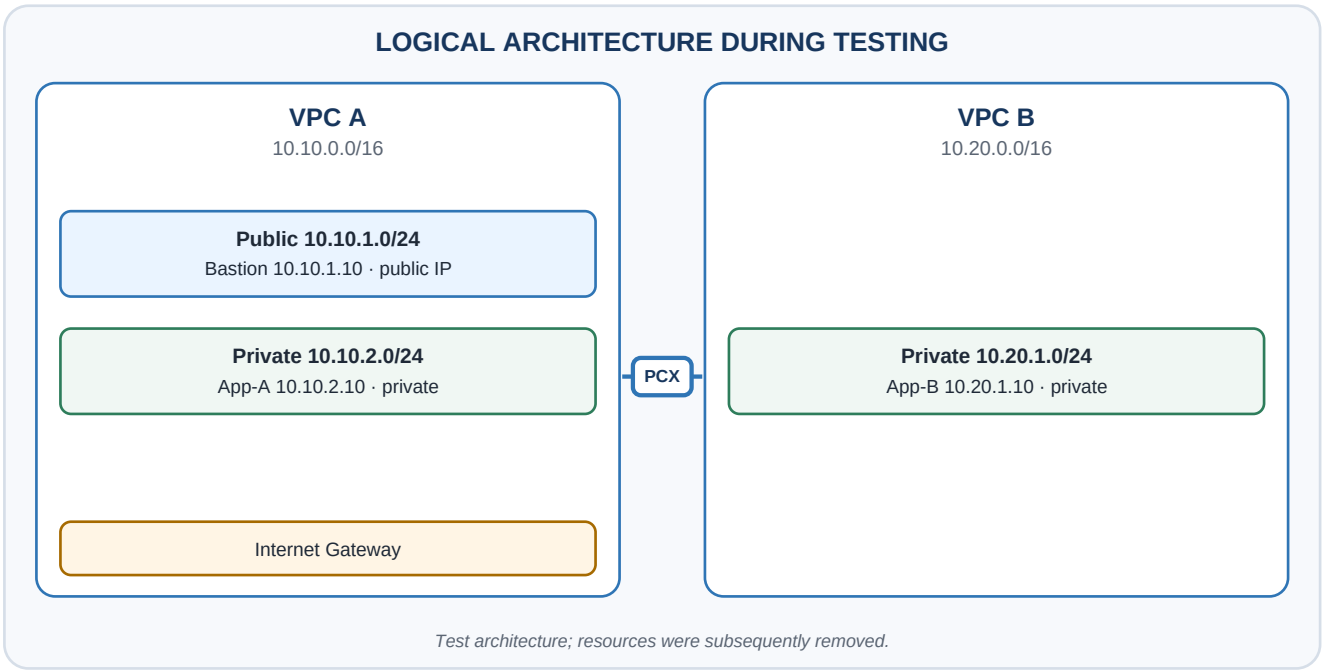
Network design, connectivity validation, controlled failure and cleanup

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Batch	AWS INTERNSHIP BATCH 1.1
AWS Region	Asia Pacific (Mumbai), ap-south-1
Report date	29 October 2026

Prepared from the supplied AWS console and terminal screenshots. Configuration pages describe the live test setup; the evidence section records the subsequent cleanup.

Architecture Overview

The lab connects two non-overlapping VPC networks through a VPC peering connection. A bastion host in VPC A provides SSH access to the private application instances. App-A and App-B have private IP addresses and no public IPv4 addresses.



Traffic path: operator → bastion → private application host for SSH; App-A ↔ App-B over VPC peering for ICMP validation.

The diagram represents the configuration during testing. The peering connection and lab resources were deleted after tests were completed.

1. Executive Summary

This project demonstrates private IPv4 connectivity between two Amazon Virtual Private Clouds (VPCs) using VPC peering. VPC A contains a public bastion host and a private application instance. VPC B contains a second private application instance. Route tables direct traffic between the VPC CIDR ranges through the peering connection, while security groups restrict inbound SSH and ICMP traffic.

Terminal evidence shows successful ping tests from App-A to App-B and from App-B to App-A, SSH access to App-B through the bastion, and a controlled ICMP failure followed by recovery after the relevant security-group rule was restored.

2. Project Objectives

- Create two non-overlapping VPCs in the Mumbai region.
- Place the bastion in VPC A's public subnet and application instances in private subnets.
- Configure VPC peering and routing in both directions.
- Limit inbound SSH and ICMP traffic with security-group rules.
- Validate forward and reverse ping, bastion-mediated SSH, and controlled failure/recovery.
- Remove lab resources after testing and preserve cleanup evidence.

3. Network Inventory

The address plan uses separate IPv4 CIDR ranges. App-A and App-B were configured with private IP addresses and no public IPv4 addresses; the bastion provided the controlled SSH entry point.

Resource	Name	CIDR / IP	Role
VPC A	ewb-vpc-a	10.10.0.0/16	Bastion + App-A
Public subnet	ewb-a-public	10.10.1.0/24	Bastion subnet
Private subnet	ewb-a-private	10.10.2.0/24	App-A subnet
VPC B	ewb-vpc-b	10.20.0.0/16	App-B network
Private subnet	ewb-b-private	10.20.1.0/24	App-B subnet
Bastion	ewb-bastion	10.10.1.10	Public SSH gateway
App-A	ewb-app-a	10.10.2.10	Private workload
App-B	ewb-app-b	10.20.1.10	Private workload

Resource IDs are retained in the original console screenshots and are not retyped where they cannot be confidently read.

4. Routing and Peering

During testing, the peering connection was active. Each route table had a route for the remote VPC CIDR targeting the peering connection. The public route table in VPC A also used a default route to the Internet Gateway.

Route table	Routes during testing
ewb-rt-a-public	10.10.0.0/16 → local 0.0.0.0/0 → Internet Gateway 10.20.0.0/16 → peering
ewb-rt-a-private	10.10.0.0/16 → local 10.20.0.0/16 → peering
ewb-rt-b-private	10.20.0.0/16 → local 10.10.0.0/16 → peering

The peering connection and its routes were removed after testing. Screenshots of the active connection belong to the test phase; screenshots showing missing peering routes belong to cleanup.

5. Security Groups

Inbound rules allowed SSH to the bastion from the operator's public IPv4 /32, SSH to private hosts from the bastion's private address, and ICMP between private workloads for connectivity testing.

Group	Inbound access	Purpose
ewb-sg-bastion	SSH TCP 22 from operator public IPv4 /32	Restricted bastion login
ewb-sg-app-a	SSH TCP 22 from 10.10.1.10/32; ICMP from 10.20.1.10/32	Private administration and ICMP
ewb-sg-app-b	SSH TCP 22 from 10.10.1.10/32; ICMP from 10.10.2.10/32	Bastion SSH and App-A ping

6. Test Results

Test	Observed evidence	Meaning
App-A → App-B ping	4 received; 0% loss	Forward and return paths worked
App-B → App-A ping	4 received; 0% loss	Reverse connectivity worked
SSH to App-B via bastion	SSH session established through bastion	Private host administrable without public IP
Remove App-B ICMP rule	Ping failed during controlled test	Inbound rule affected reachability
Restore App-B ICMP rule	Ping succeeded after restoration	Connectivity recovered

7. Cleanup Summary

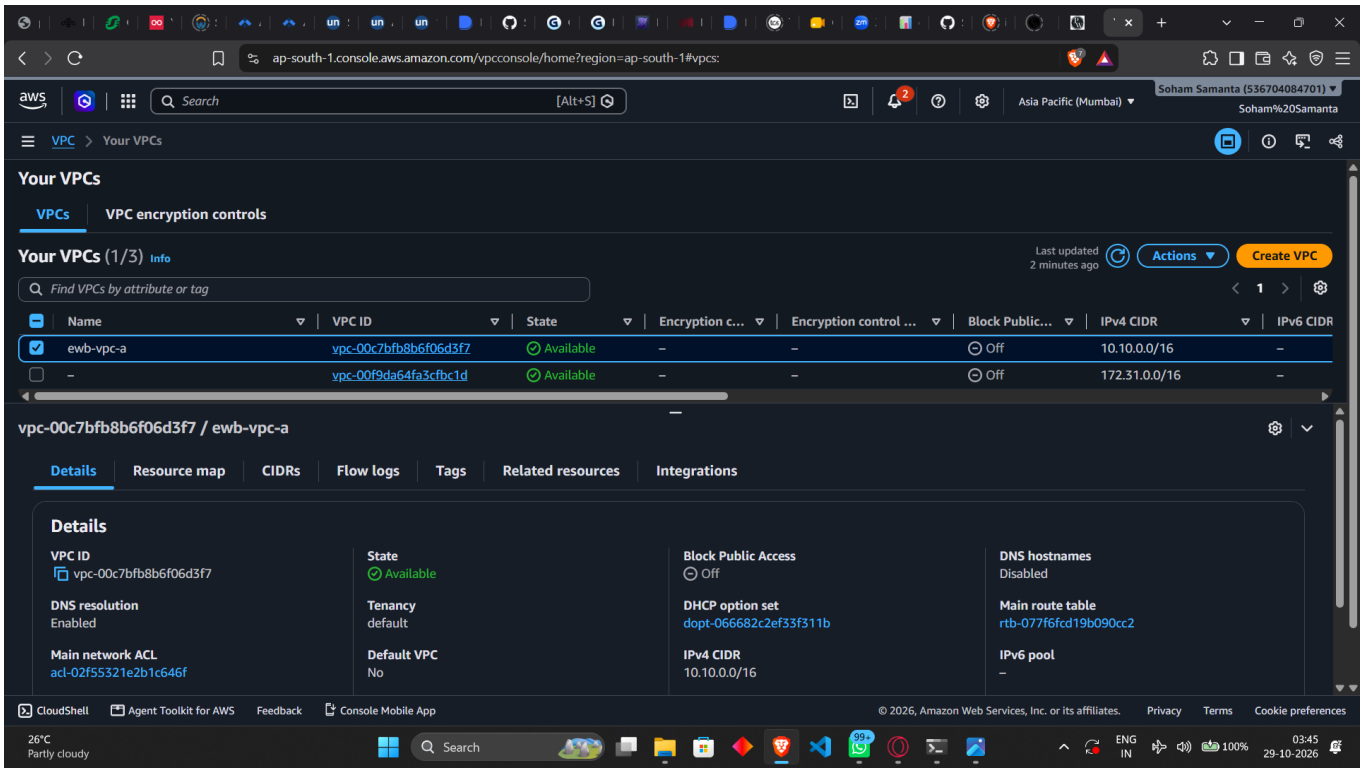
- All three EC2 instances terminated.
- EBS volumes and snapshots lists cleared.
- VPC peering connection and peering routes removed.
- All three lab security groups deleted.
- All three lab subnets and custom route tables deleted.
- Internet Gateway detached and deleted.
- **Both lab VPCs were successfully removed.** The supplied final VPC screenshot shows the lab VPCs are no longer present in the displayed list.

8. Screenshot Evidence Register

The screenshots below are preserved as supplied, scaled proportionally on portrait pages, and accompanied by concise captions. Configuration and test screenshots show the live lab setup; cleanup screenshots show the subsequent resource removal.

Figure 1. VPC A configuration

Shows VPC A's console configuration and CIDR.

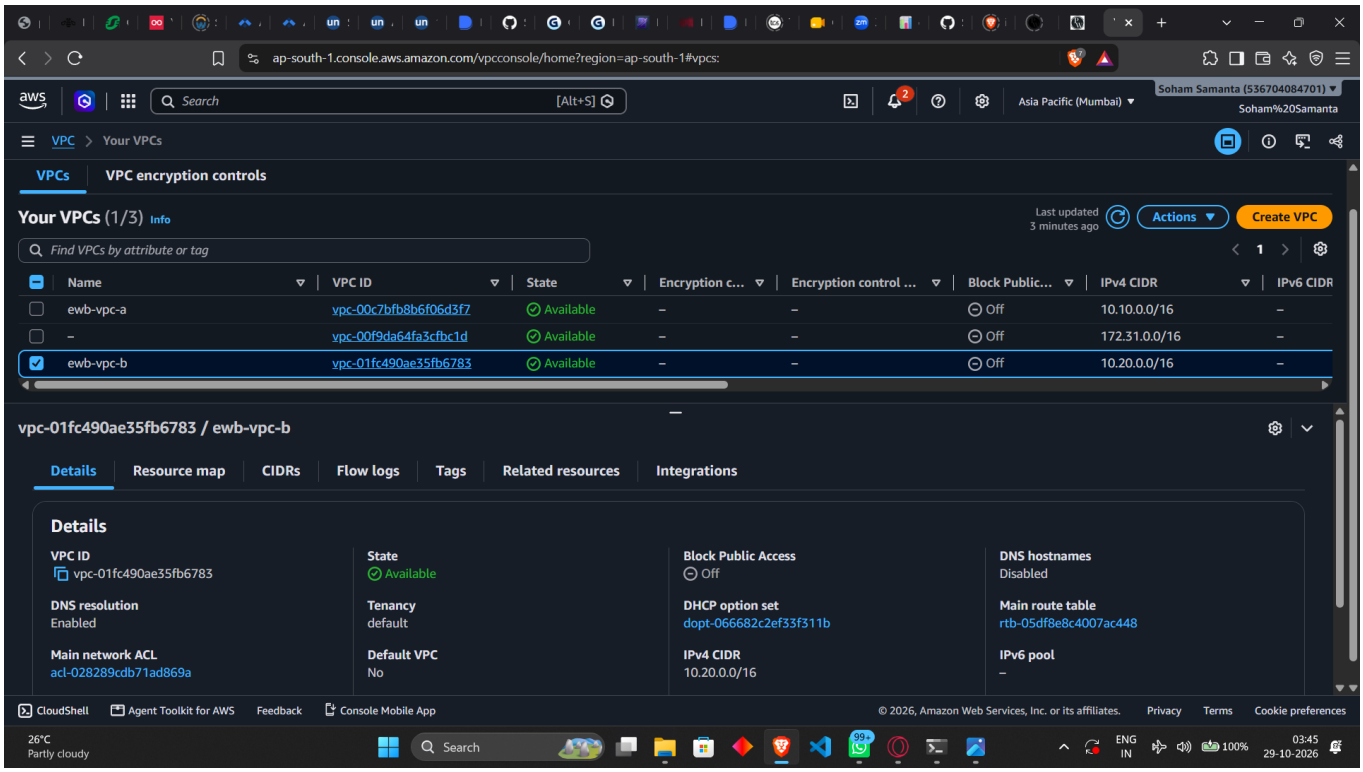


EVIDENCE PURPOSE	HOW TO READ IT
Shows VPC A's console configuration and CIDR.	This figure supports the stage identified in the caption. It should be read with the configuration and test summaries; the report does not infer hidden values from unreadable areas.

Original screenshot: VPC A.png

Figure 2. VPC B configuration

Shows VPC B's console configuration and CIDR.

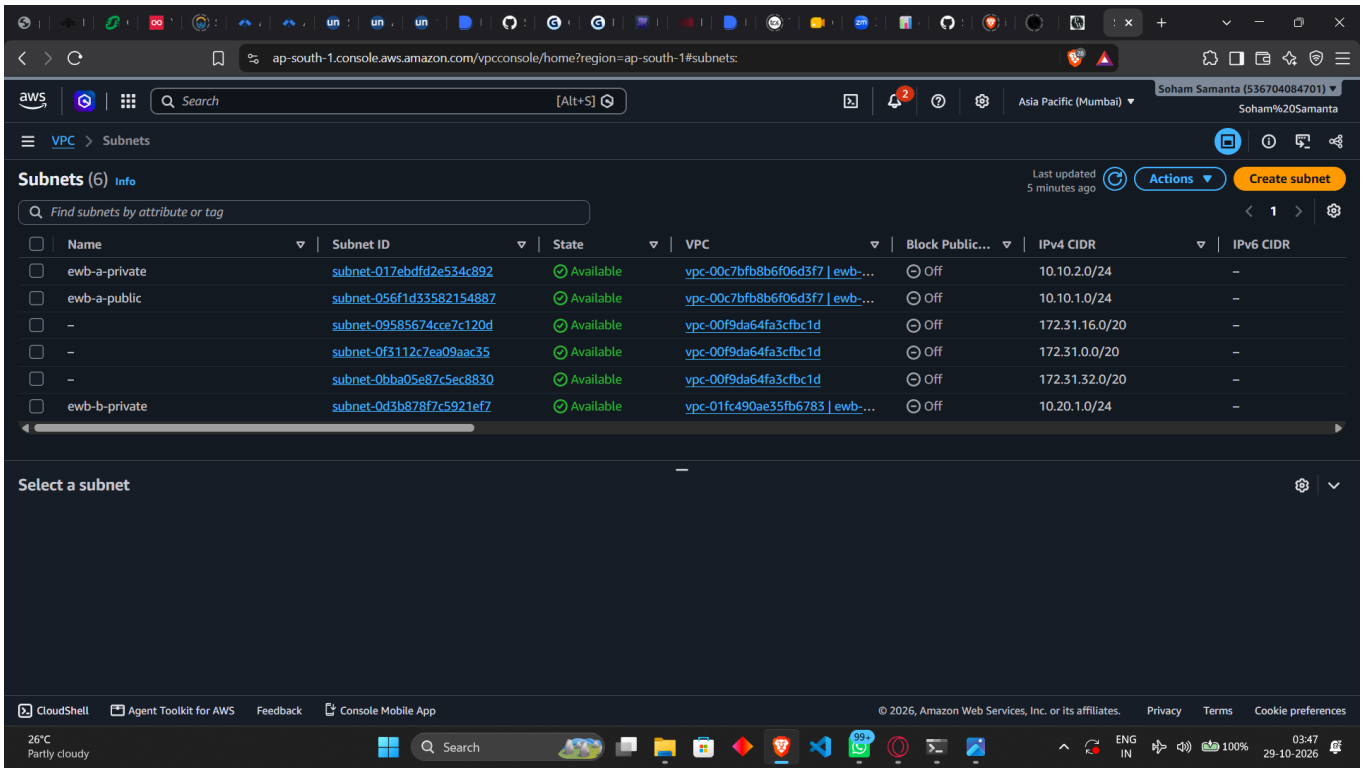


EVIDENCE PURPOSE	HOW TO READ IT
Shows VPC B's console configuration and CIDR.	This figure supports the stage identified in the caption. It should be read with the configuration and test summaries; the report does not infer hidden values from unreadable areas.

Original screenshot: VPC B.png

Figure 3. Subnet inventory before cleanup

Shows the three lab subnets and their CIDR allocations.

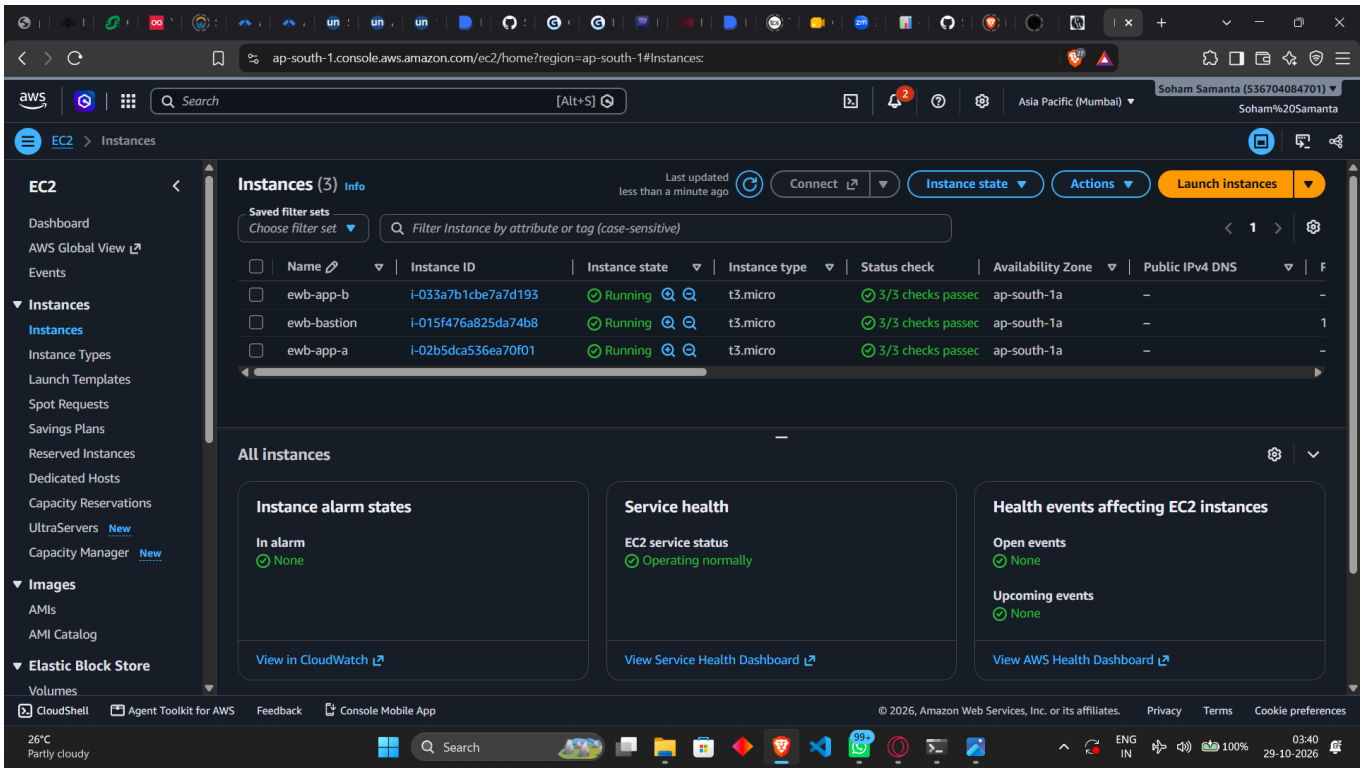


EVIDENCE PURPOSE	HOW TO READ IT
Shows the three lab subnets and their CIDR allocations.	This figure supports the stage identified in the caption. It should be read with the configuration and test summaries; the report does not infer hidden values from unreadable areas.

Original screenshot: SUBNETS.png

Figure 4. EC2 instance inventory

Shows the three instances during the lab; App-A and App-B were private-only.

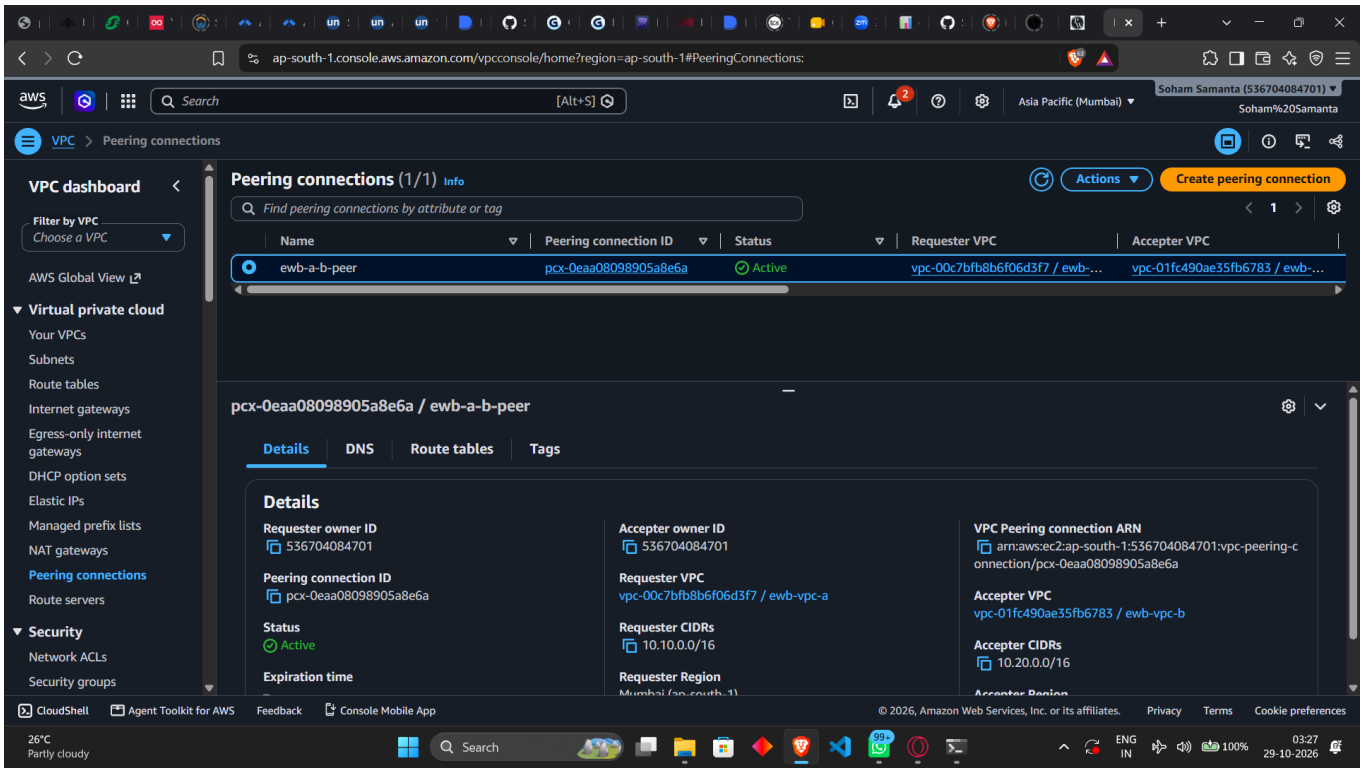


EVIDENCE PURPOSE	HOW TO READ IT
Shows the three instances during the lab; App-A and App-B were private-only.	This figure supports the stage identified in the caption. It should be read with the configuration and test summaries; the report does not infer hidden values from unreadable areas.

Original screenshot: INSTANCES.png

Figure 5. Active VPC peering connection

Shows the peering connection during testing.

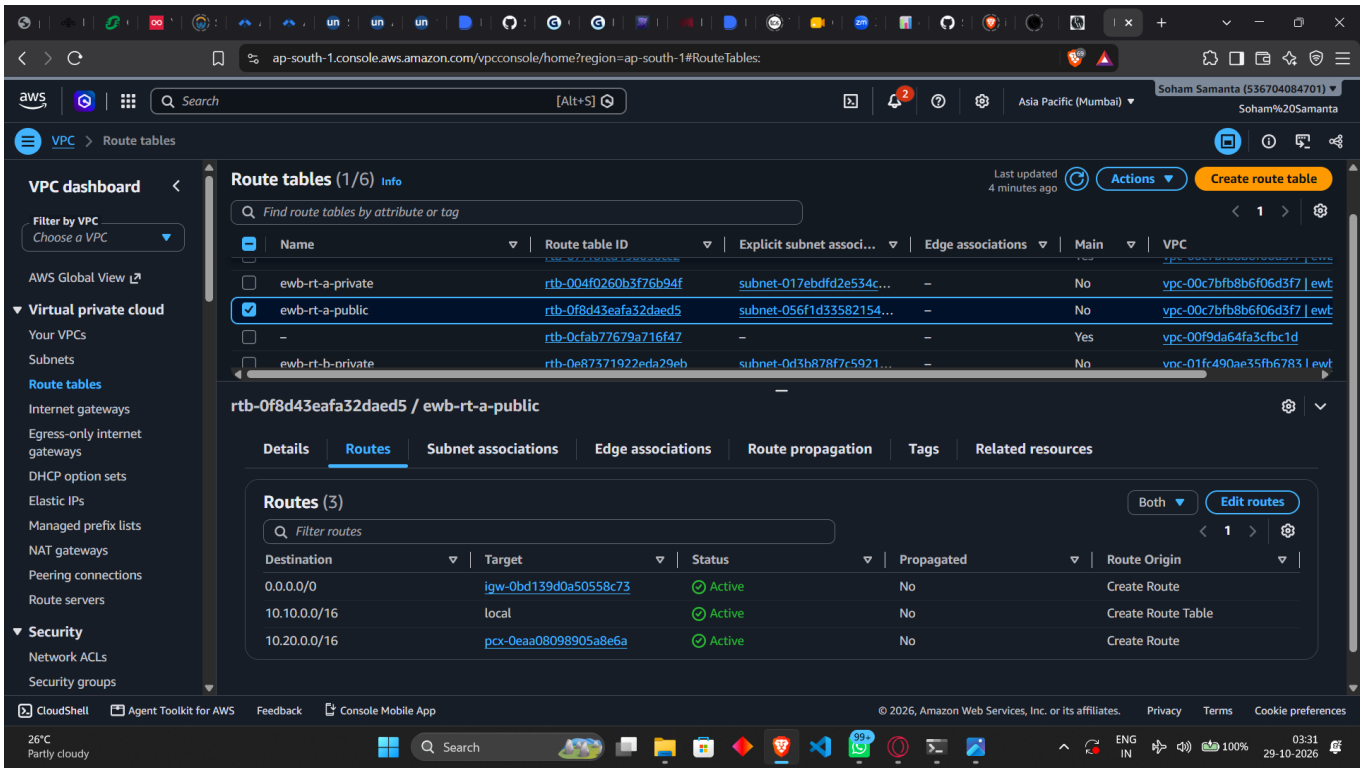


EVIDENCE PURPOSE	HOW TO READ IT
Shows the peering connection during testing.	This figure supports the stage identified in the caption. It should be read with the configuration and test summaries; the report does not infer hidden values from unreadable areas.

Original screenshot: peering_connection.png

Figure 6. Public route table during testing

Shows VPC A public route table configuration.

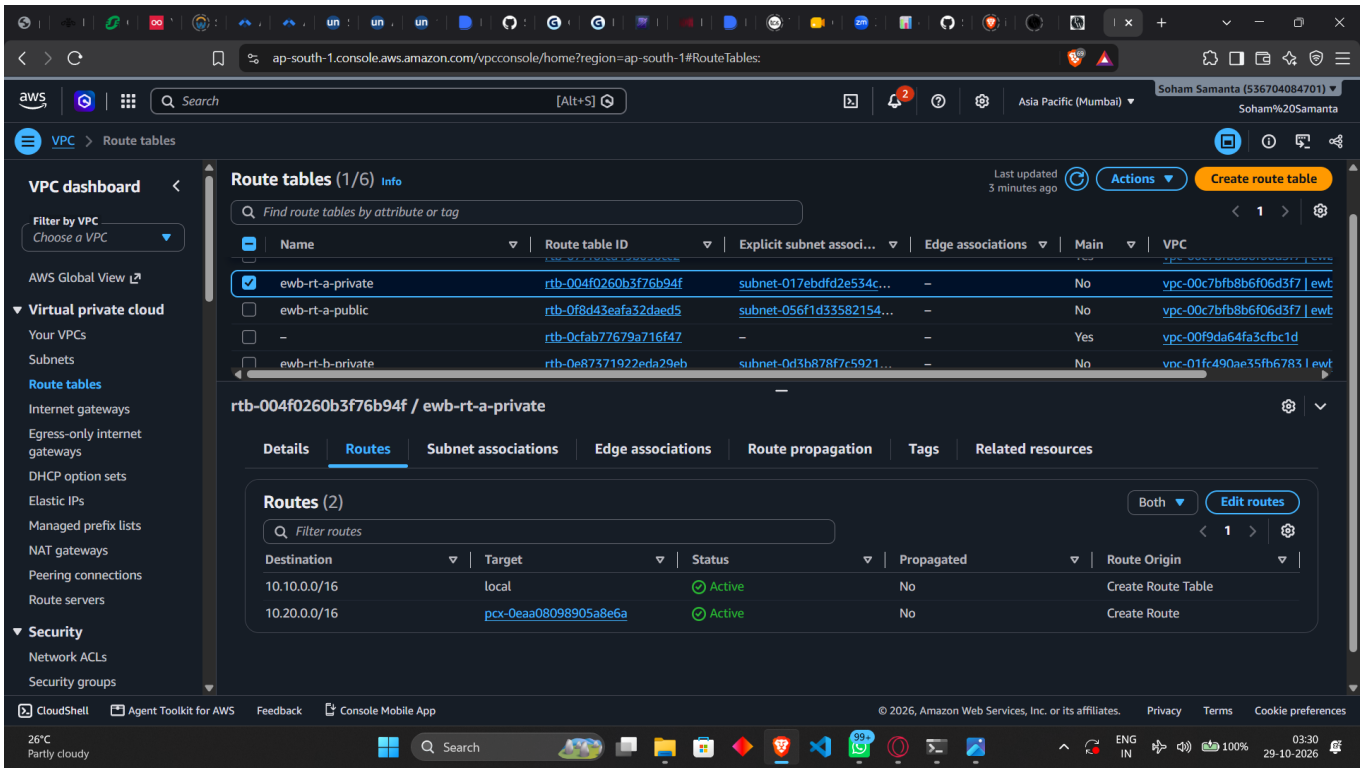


EVIDENCE PURPOSE	HOW TO READ IT
Shows VPC A public route table configuration.	This figure supports the stage identified in the caption. It should be read with the configuration and test summaries; the report does not infer hidden values from unreadable areas.

Original screenshot: route a public.png

Figure 7. VPC A private route table during testing

Shows VPC A private routing configuration.

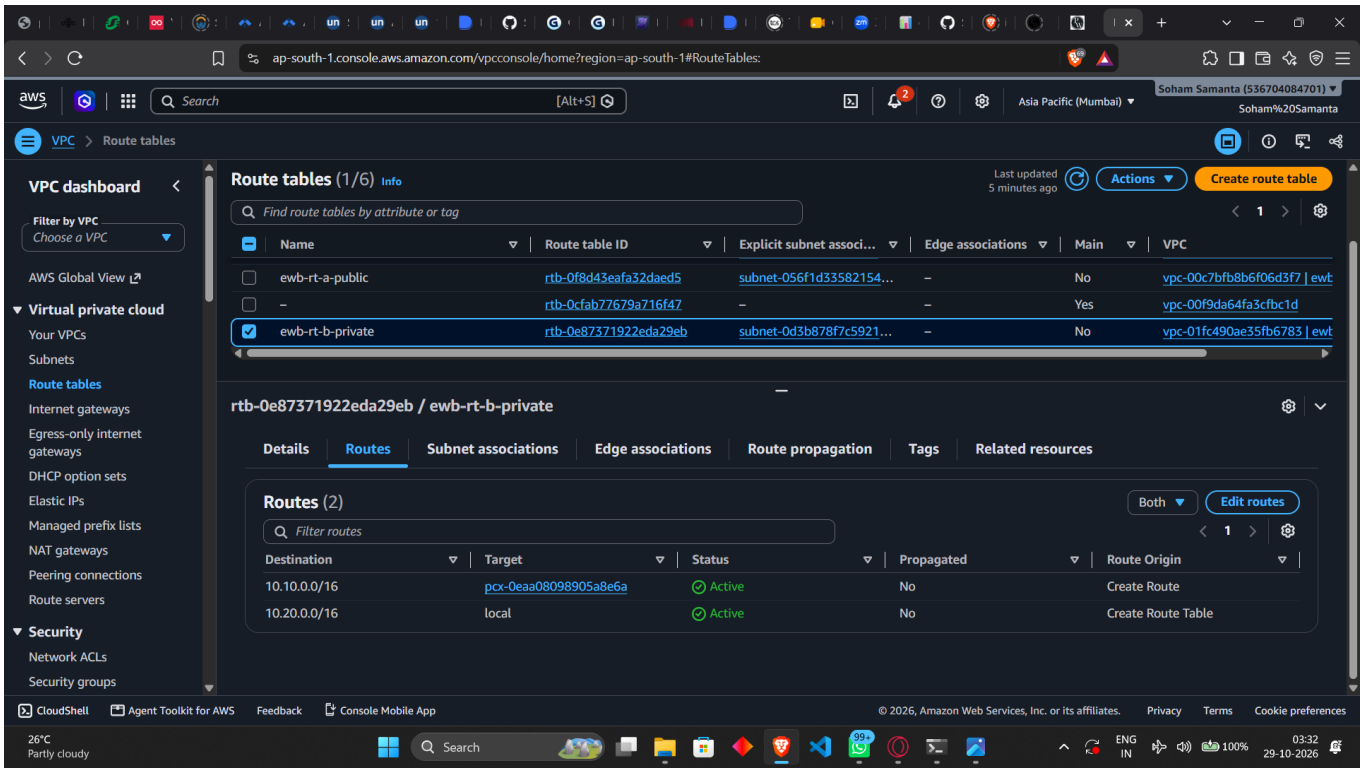


EVIDENCE PURPOSE	HOW TO READ IT
Shows VPC A private routing configuration.	This figure supports the stage identified in the caption. It should be read with the configuration and test summaries; the report does not infer hidden values from unreadable areas.

Original screenshot: route a private.png

Figure 8. VPC B private route table during testing

Shows VPC B routing configuration.

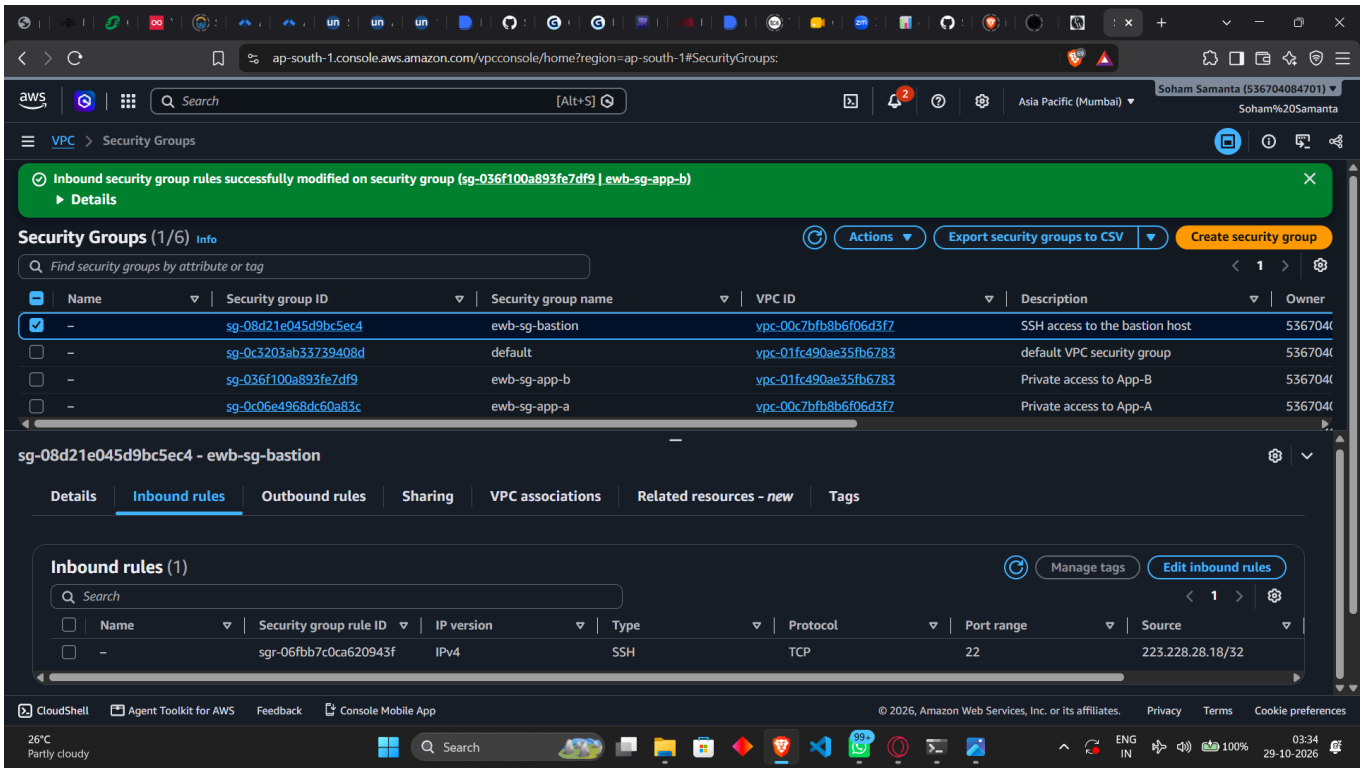


EVIDENCE PURPOSE	HOW TO READ IT
Shows VPC B routing configuration.	This figure supports the stage identified in the caption. It should be read with the configuration and test summaries; the report does not infer hidden values from unreadable areas.

Original screenshot: route b pvt.png

Figure 9. Bastion security group

Shows the bastion inbound access rule.

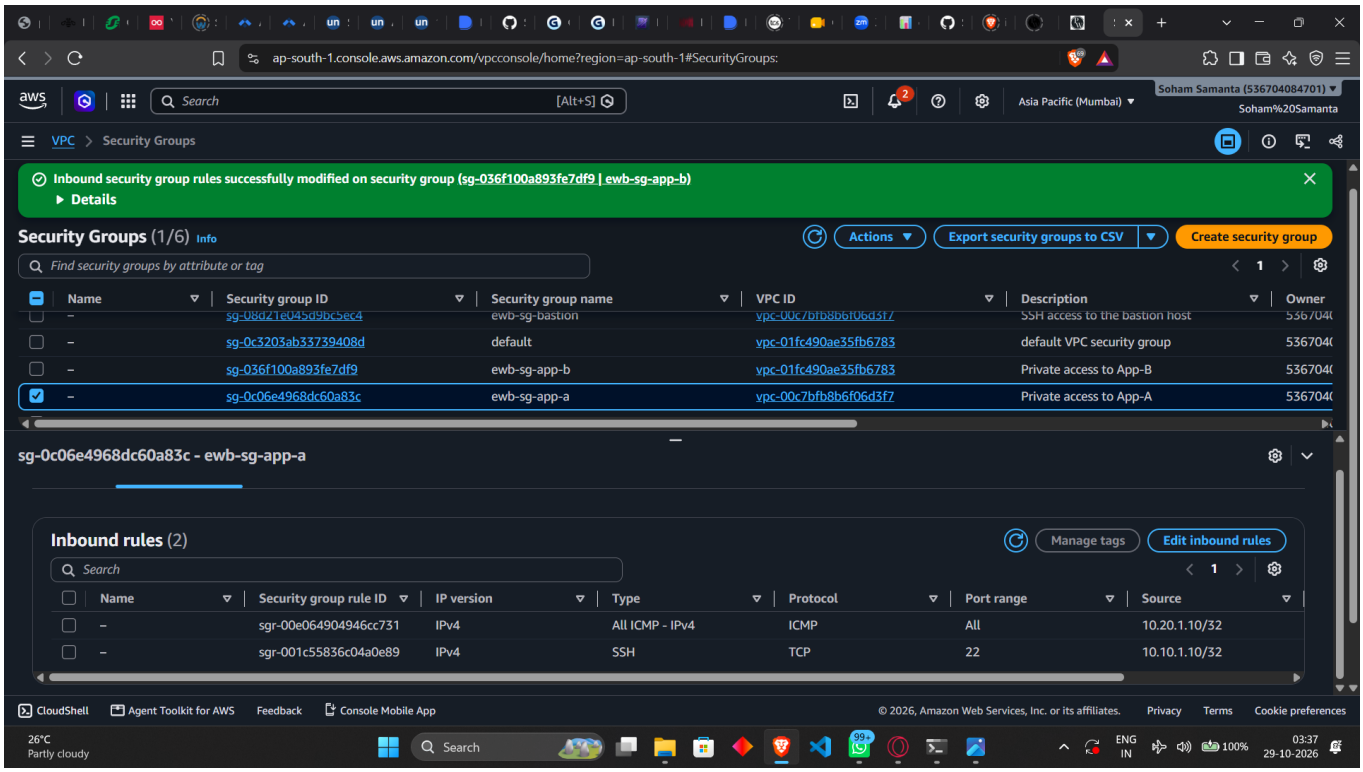


EVIDENCE PURPOSE	HOW TO READ IT
Shows the bastion inbound access rule.	This figure supports the stage identified in the caption. It should be read with the configuration and test summaries; the report does not infer hidden values from unreadable areas.

Original screenshot: bastion rule.png

Figure 10. App-A security group

Shows App-A inbound rules.

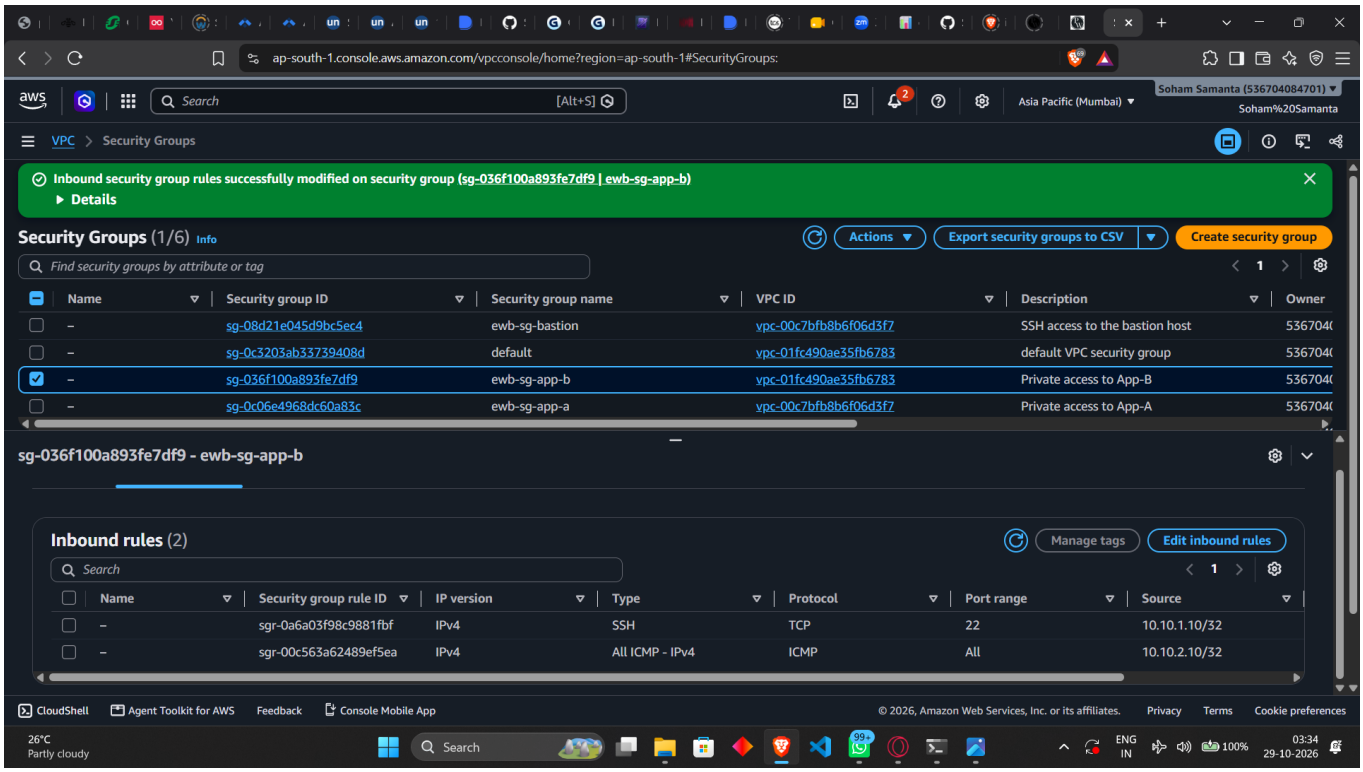


EVIDENCE PURPOSE	HOW TO READ IT
Shows App-A inbound rules.	This figure supports the stage identified in the caption. It should be read with the configuration and test summaries; the report does not infer hidden values from unreadable areas.

Original screenshot: app a rules.png

Figure 11. App-B security group

Shows App-B inbound rules used during tests.



EVIDENCE PURPOSE	HOW TO READ IT
Shows App-B inbound rules used during tests.	This figure supports the stage identified in the caption. It should be read with the configuration and test summaries; the report does not infer hidden values from unreadable areas.

Original screenshot: app b rules.png

Figure 12. Baseline Connectivity Test: App-A to App-B

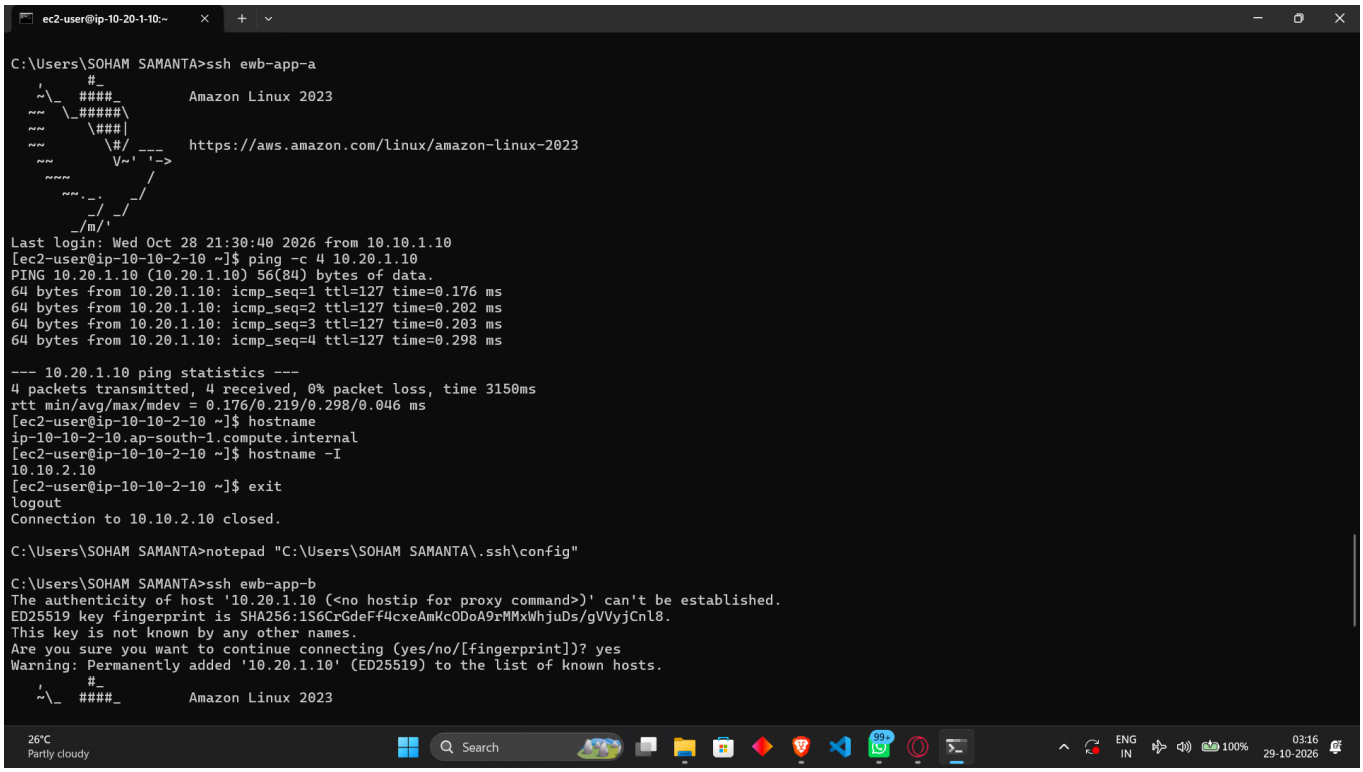
Ping from App-A to App-B before the VPC peering routes were configured

Description

Running `ping -c 4 10.20.1.10` from App-A before the peering routes were configured gives no replies, because traffic to the remote VPC does not yet have a peering route. Apologies, the screenshot for this test was not captured.

Figure 13. App-A host and forward ping

Terminal evidence for App-A identity and the forward ping test.

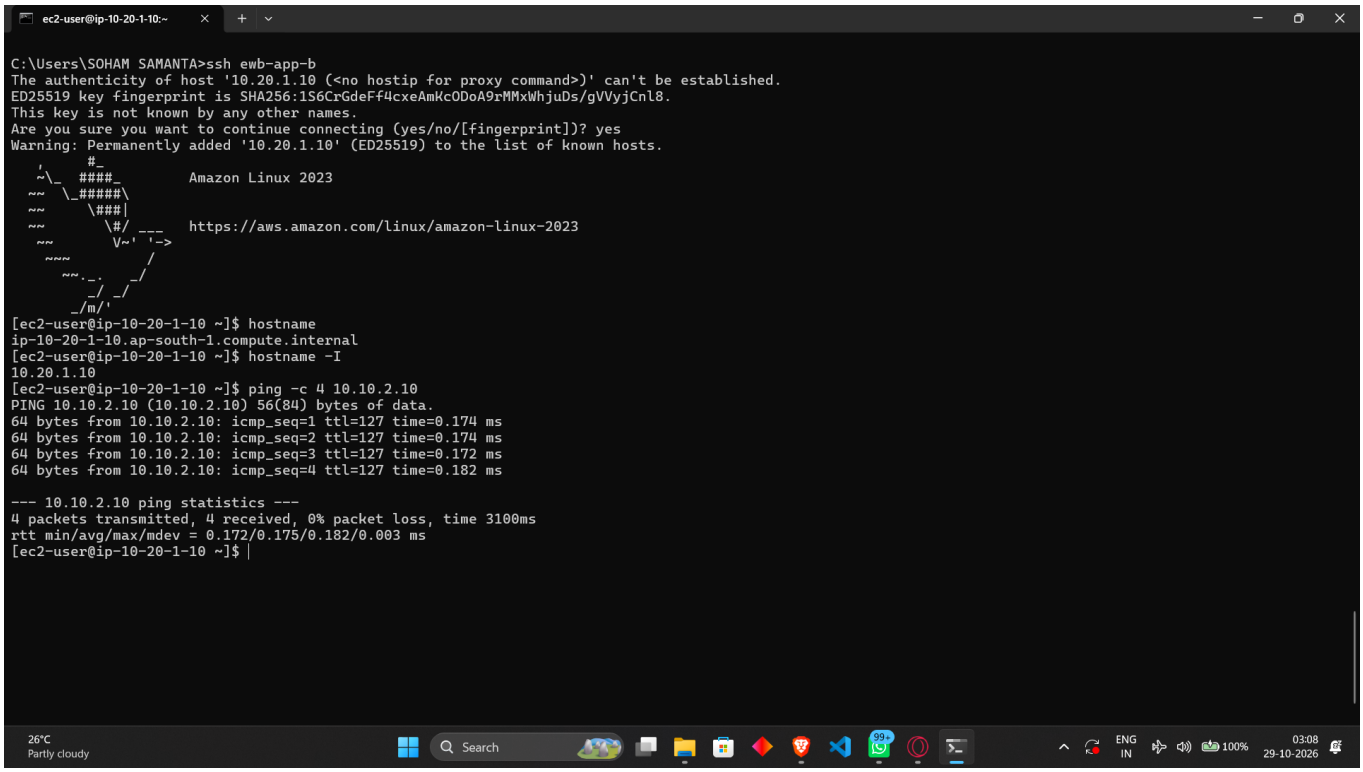


EVIDENCE PURPOSE	HOW TO READ IT
Terminal evidence for App-A identity and the forward ping test.	This figure supports the stage identified in the caption. It should be read with the configuration and test summaries; the report does not infer hidden values from unreadable areas.

Original screenshot: ewb-app-a connection and test.png

Figure 14. App-B host and reverse ping

Terminal evidence for App-B identity and the reverse ping test.

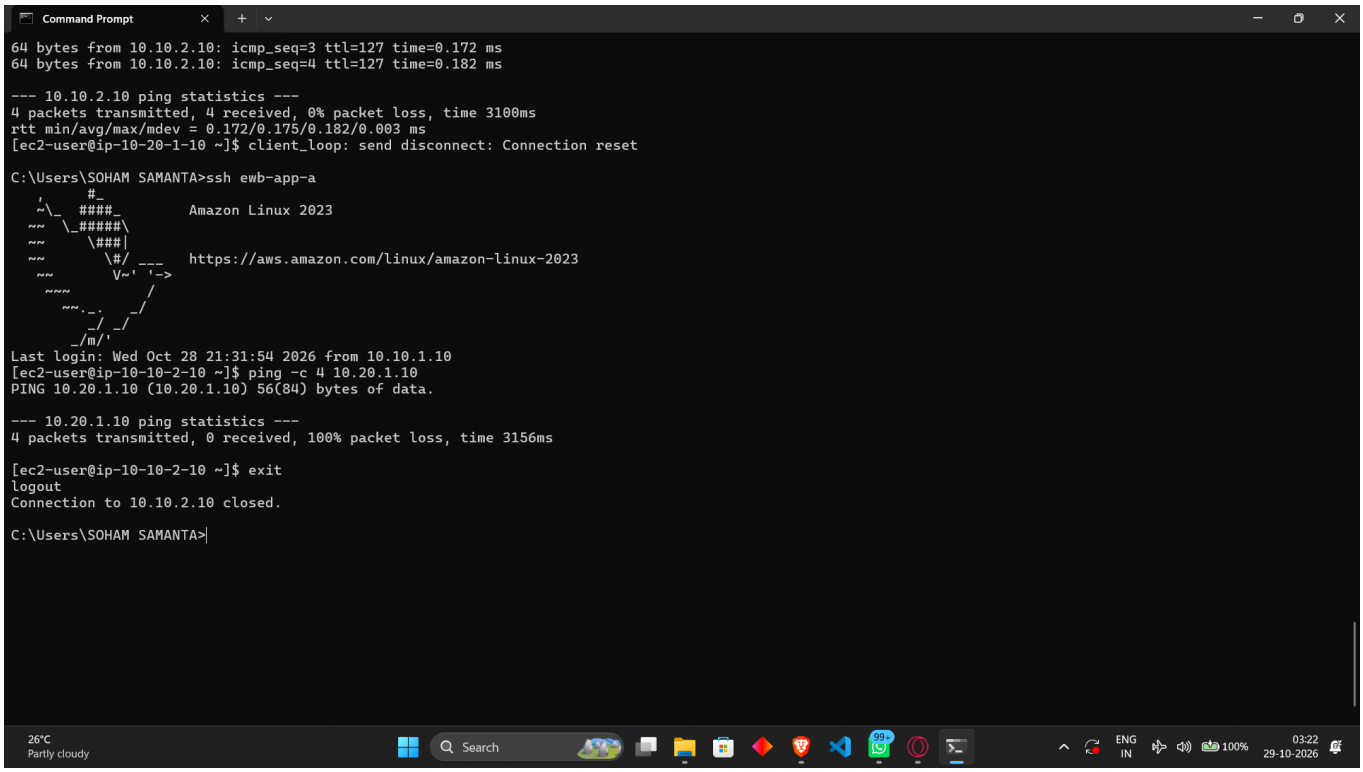


EVIDENCE PURPOSE	HOW TO READ IT
Terminal evidence for App-B identity and the reverse ping test.	This figure supports the stage identified in the caption. It should be read with the configuration and test summaries; the report does not infer hidden values from unreadable areas.

Original screenshot: ewb-app-b connectionand test.png

Figure 15. Controlled ICMP failure

Shows failed ping while App-B's inbound ICMP rule was removed.

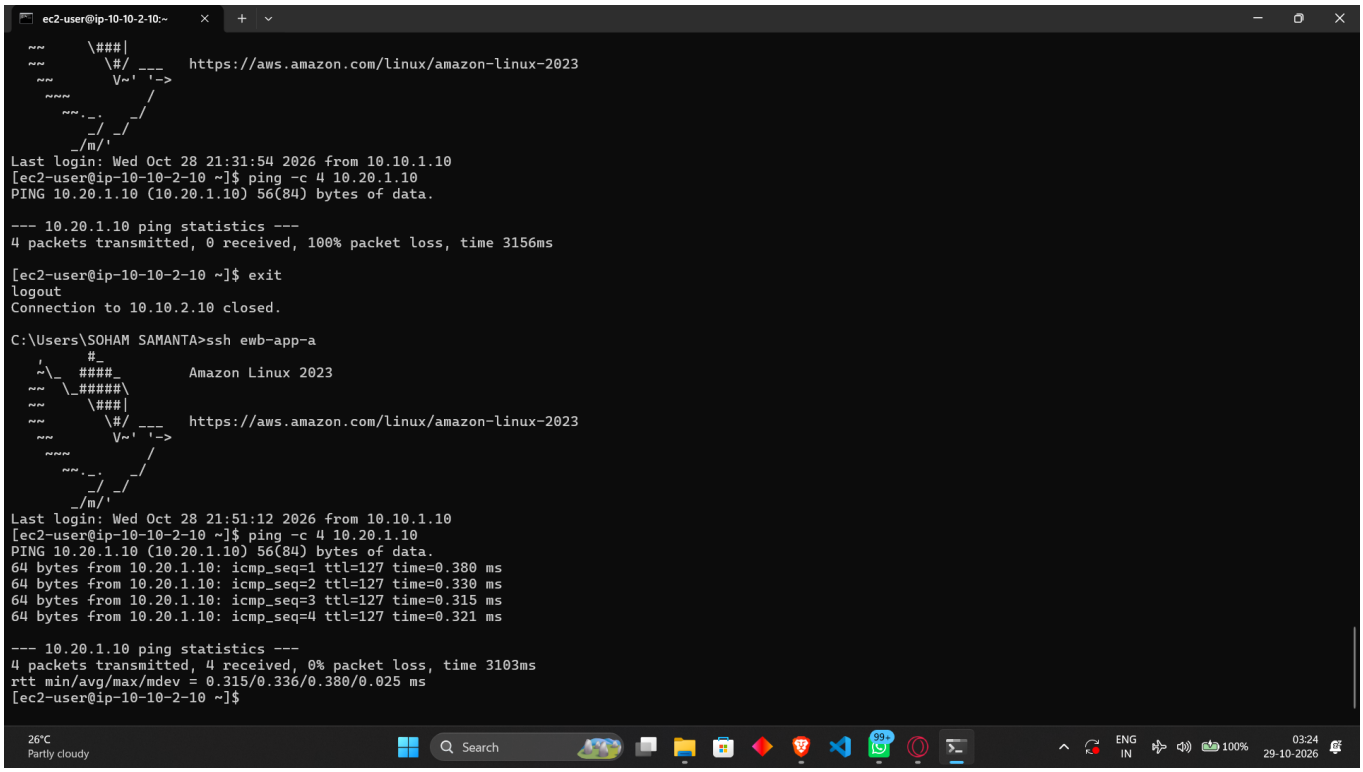


EVIDENCE PURPOSE	HOW TO READ IT
Shows failed ping while App-B's inbound ICMP rule was removed.	This figure supports the stage identified in the caption. It should be read with the configuration and test summaries; the report does not infer hidden values from unreadable areas.

Original screenshot: failure removing icmp rule.png

Figure 16. Connectivity restored

Shows ping success after the ICMP rule was restored.

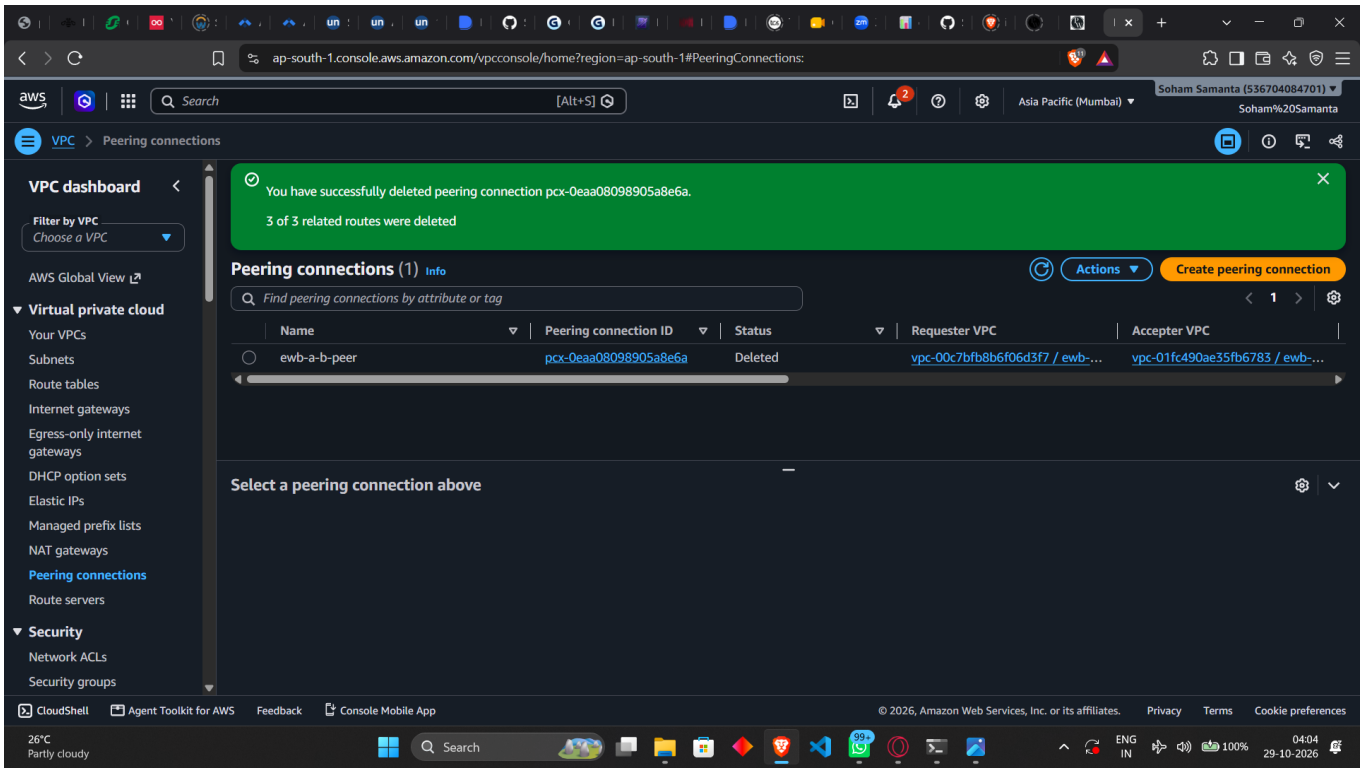


EVIDENCE PURPOSE	HOW TO READ IT
Shows ping success after the ICMP rule was restored.	This figure supports the stage identified in the caption. It should be read with the configuration and test summaries; the report does not infer hidden values from unreadable areas.

Original screenshot: success after restore.png

Figure 17. Peering connection removed

Cleanup evidence for the peering connection.

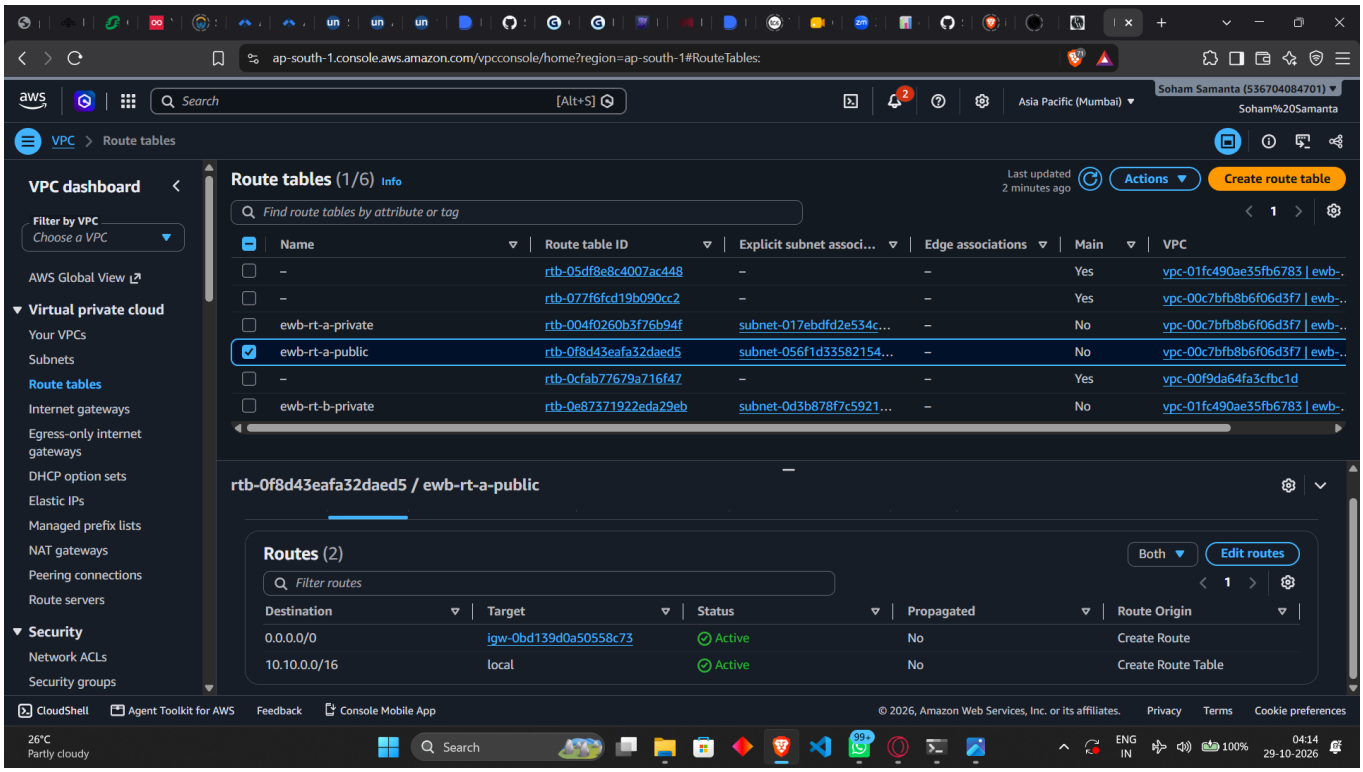


EVIDENCE PURPOSE	HOW TO READ IT
Cleanup evidence for the peering connection.	This figure supports the stage identified in the caption. It should be read with the configuration and test summaries; the report does not infer hidden values from unreadable areas.

Original screenshot: peering connection removed.png

Figure 18. Public route table after peering cleanup

Shows the peering route removed from the public table.

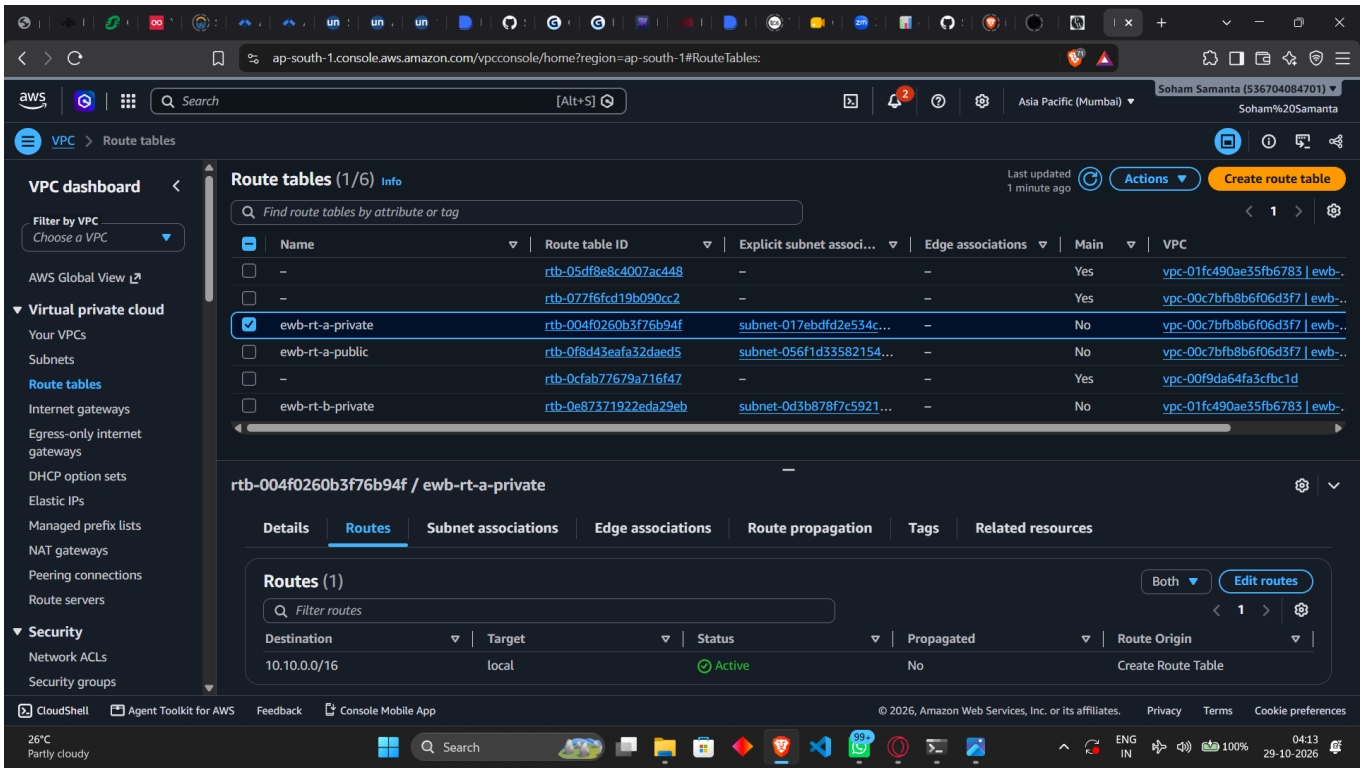


EVIDENCE PURPOSE	HOW TO READ IT
Shows the peering route removed from the public table.	This figure supports the stage identified in the caption. It should be read with the configuration and test summaries; the report does not infer hidden values from unreadable areas.

Original screenshot: a pbl routing table clear 1.png

Figure 19. VPC A private route table after cleanup

Shows the peering route removed from VPC A private table.

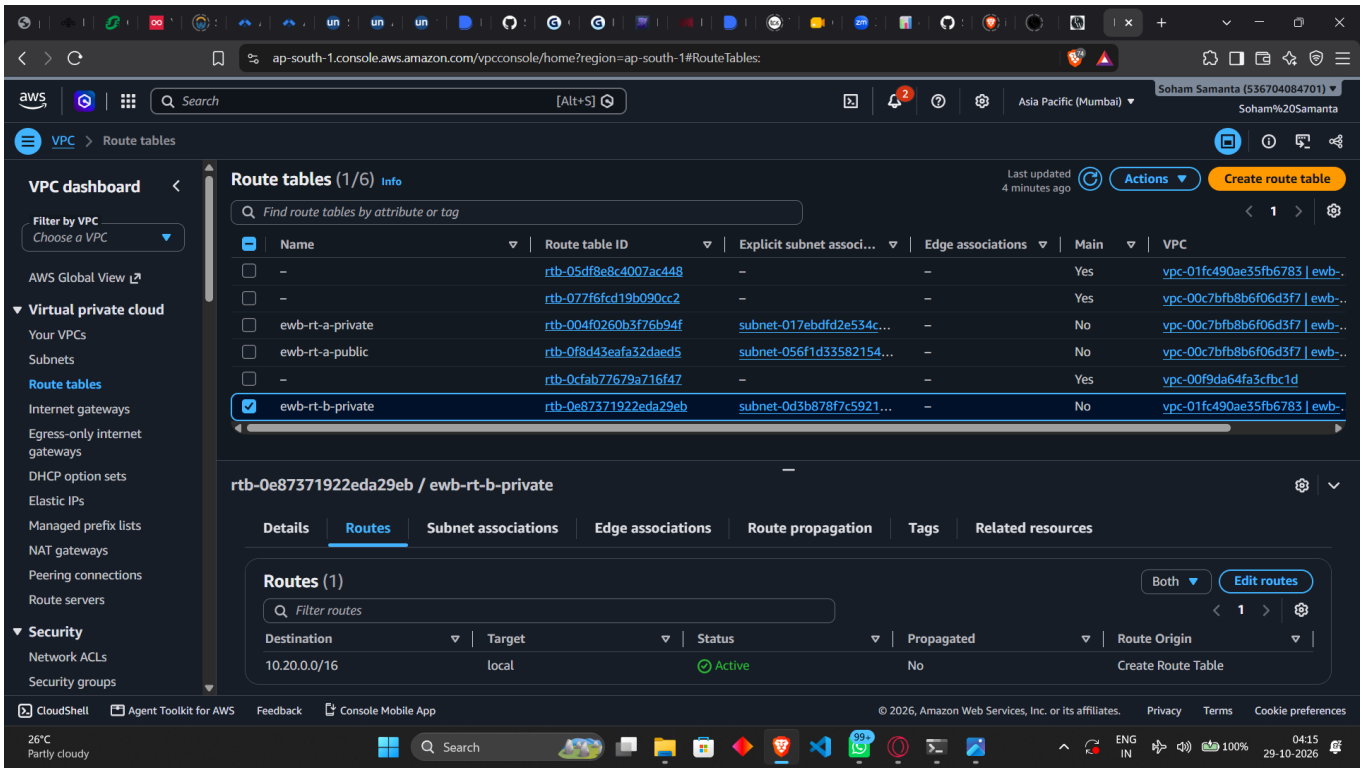


EVIDENCE PURPOSE	HOW TO READ IT
Shows the peering route removed from VPC A private table.	This figure supports the stage identified in the caption. It should be read with the configuration and test summaries; the report does not infer hidden values from unreadable areas.

Original screenshot: a pvt routing table clear 1.png

Figure 20. VPC B private route table after cleanup

Shows the peering route removed from VPC B private table.

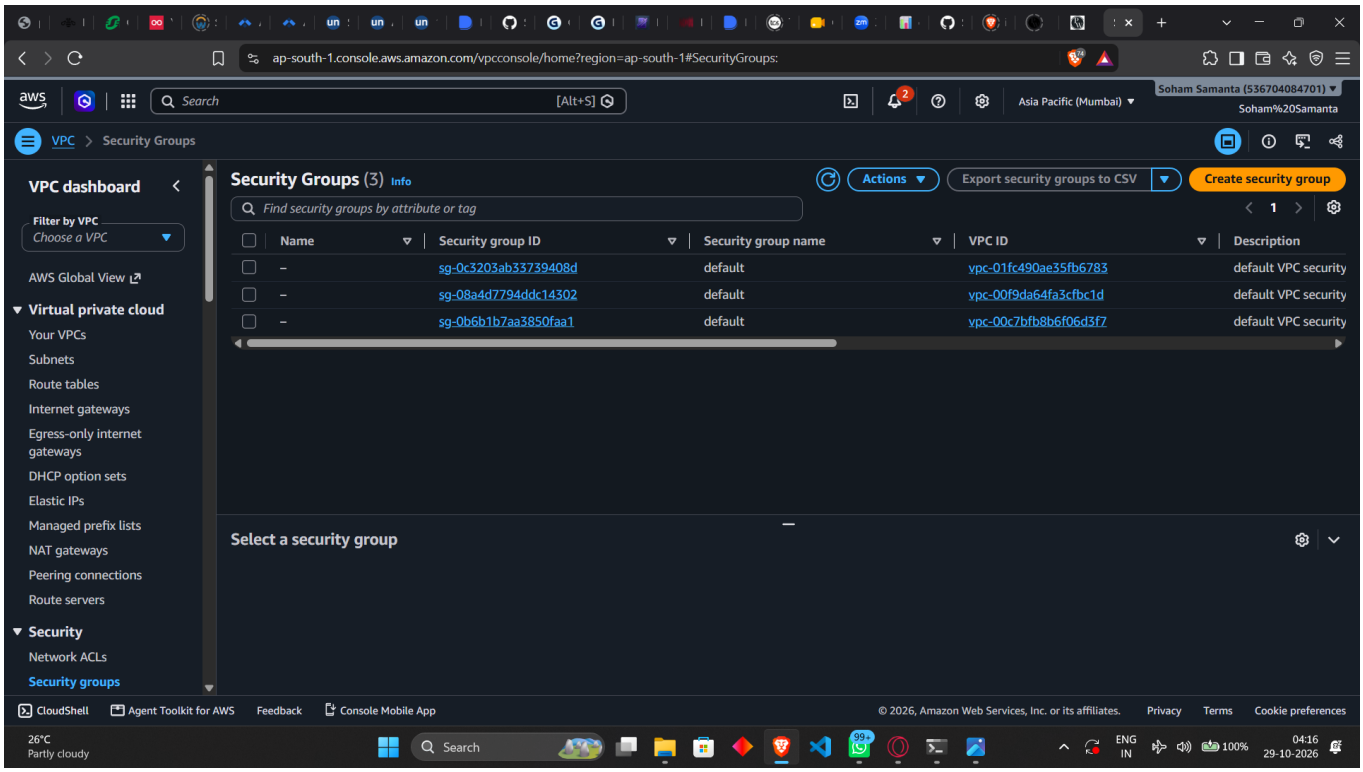


EVIDENCE PURPOSE	HOW TO READ IT
Shows the peering route removed from VPC B private table.	This figure supports the stage identified in the caption. It should be read with the configuration and test summaries; the report does not infer hidden values from unreadable areas.

Original screenshot: b pvt routing table cleared.png

Figure 21. Security groups deleted

Shows cleanup of the lab security groups.

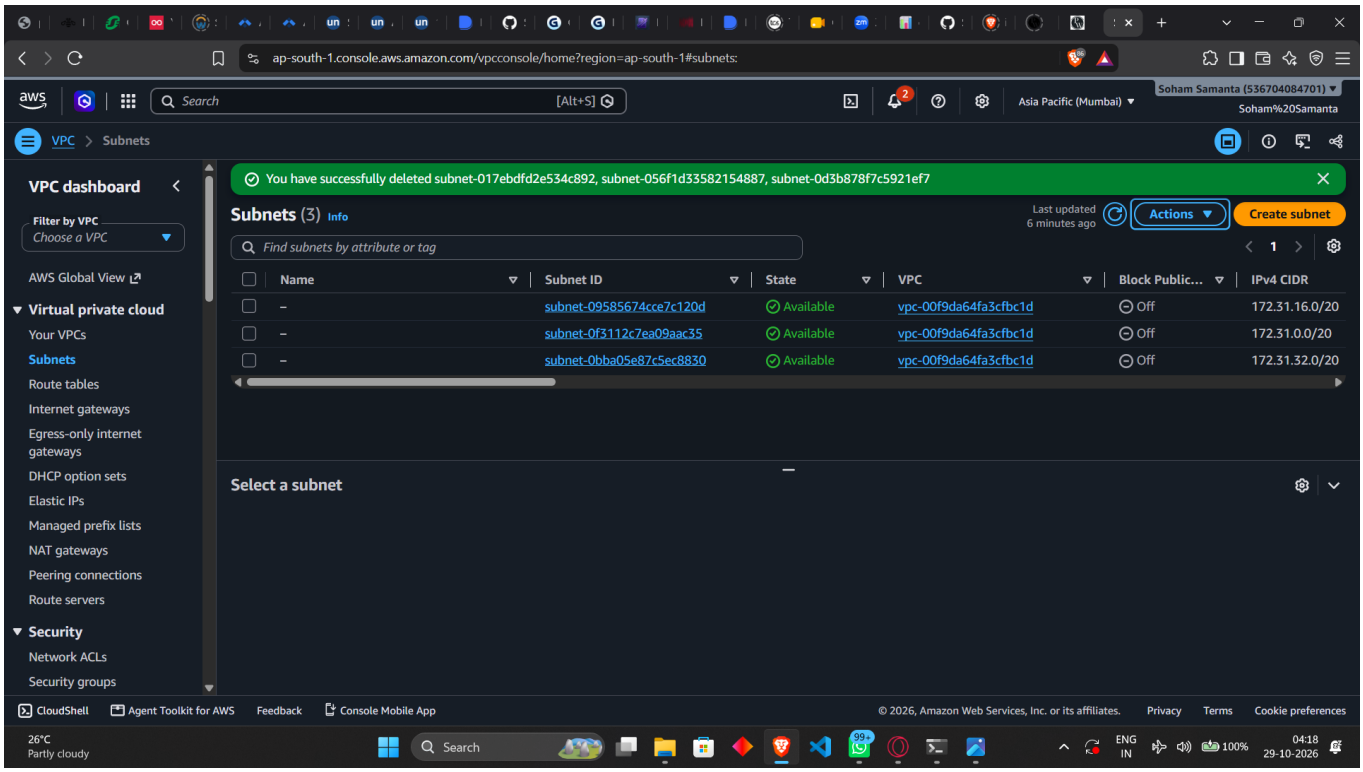


EVIDENCE PURPOSE	HOW TO READ IT
Shows cleanup of the lab security groups.	This figure supports the stage identified in the caption. It should be read with the configuration and test summaries; the report does not infer hidden values from unreadable areas.

Original screenshot: sgs deleted.png

Figure 22. Subnets deleted

Shows cleanup of the lab subnets.

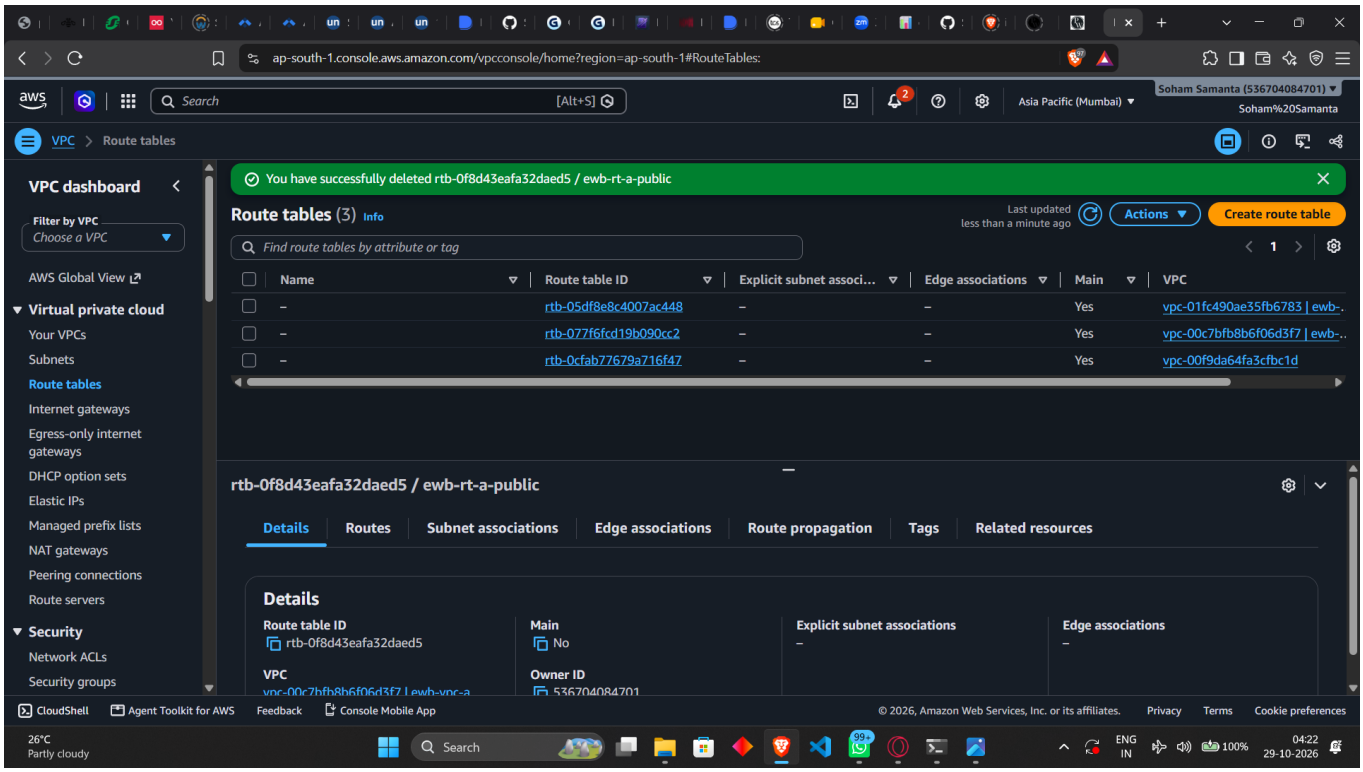


EVIDENCE PURPOSE	HOW TO READ IT
Shows cleanup of the lab subnets.	This figure supports the stage identified in the caption. It should be read with the configuration and test summaries; the report does not infer hidden values from unreadable areas.

Original screenshot: subnets deleted.png

Figure 23. Custom route tables removed

Shows cleanup of the custom route tables.

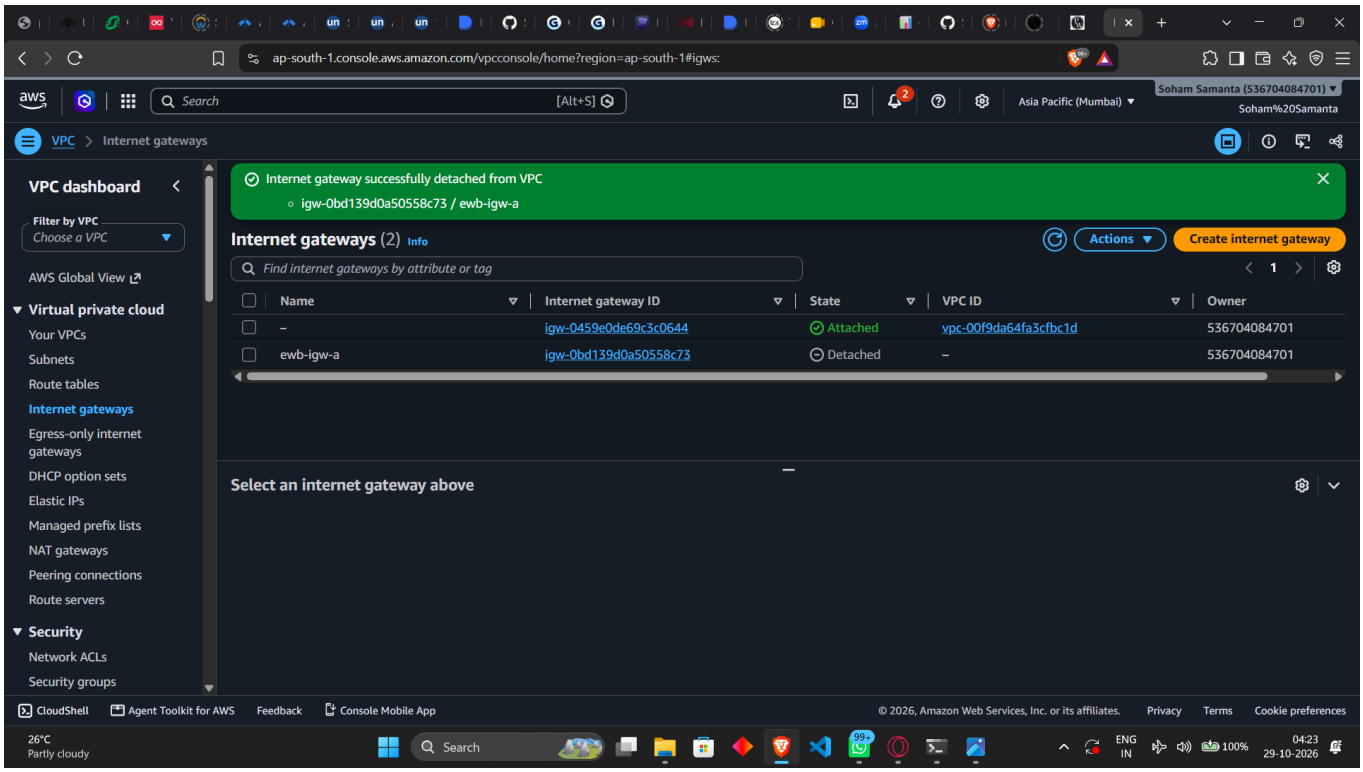


EVIDENCE PURPOSE	HOW TO READ IT
Shows cleanup of the custom route tables.	This figure supports the stage identified in the caption. It should be read with the configuration and test summaries; the report does not infer hidden values from unreadable areas.

Original screenshot: route tables removed.png

Figure 24. Internet Gateway detached

Shows the Internet Gateway detached.

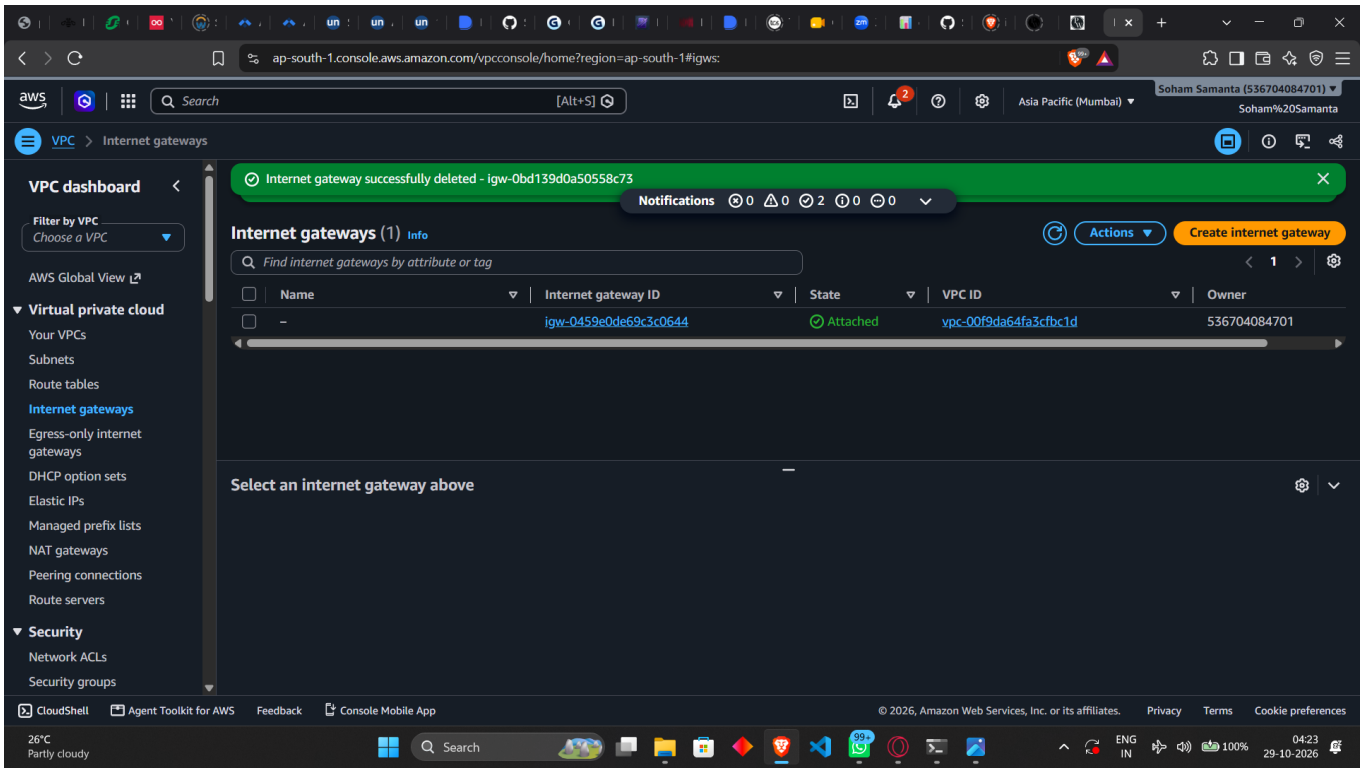


EVIDENCE PURPOSE	HOW TO READ IT
Shows the Internet Gateway detached.	This figure supports the stage identified in the caption. It should be read with the configuration and test summaries; the report does not infer hidden values from unreadable areas.

Original screenshot: igw detached.png

Figure 25. Internet Gateway deleted

Shows the Internet Gateway deletion.

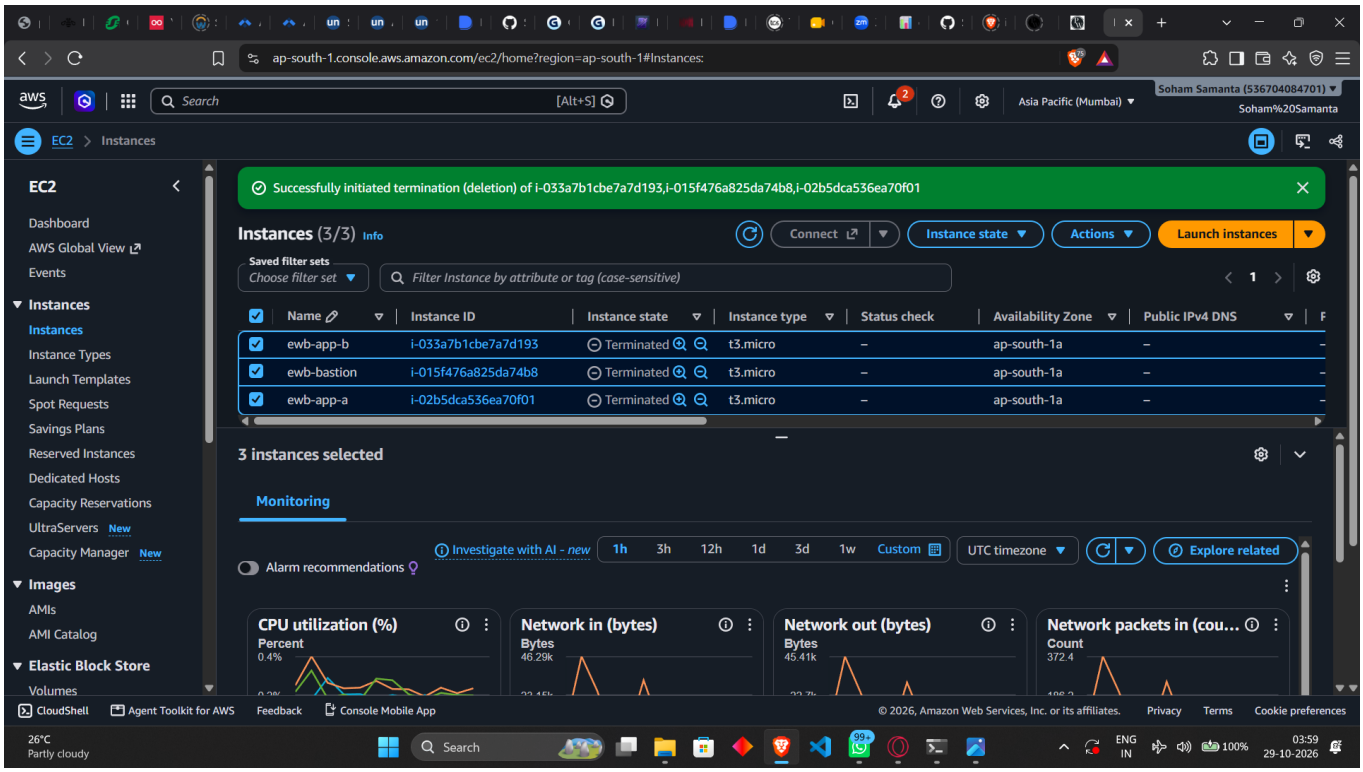


EVIDENCE PURPOSE	HOW TO READ IT
Shows the Internet Gateway deletion.	This figure supports the stage identified in the caption. It should be read with the configuration and test summaries; the report does not infer hidden values from unreadable areas.

Original screenshot: igw deleted.png

Figure 26. EC2 instances terminated

Shows the instance termination stage.

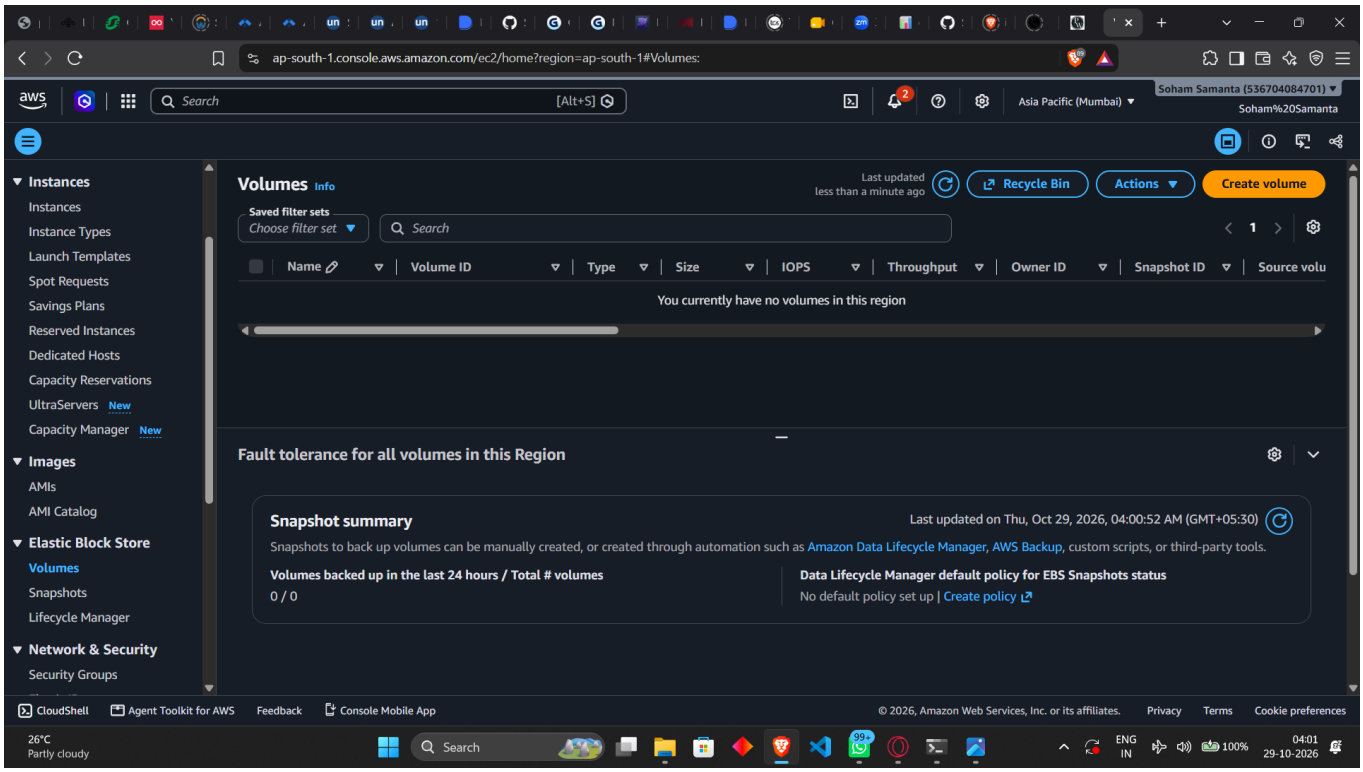


EVIDENCE PURPOSE	HOW TO READ IT
Shows the instance termination stage.	This figure supports the stage identified in the caption. It should be read with the configuration and test summaries; the report does not infer hidden values from unreadable areas.

Original screenshot: EC2 TERMINATED.png

Figure 27. EBS volumes cleared

Shows no lab EBS volumes in the displayed list.

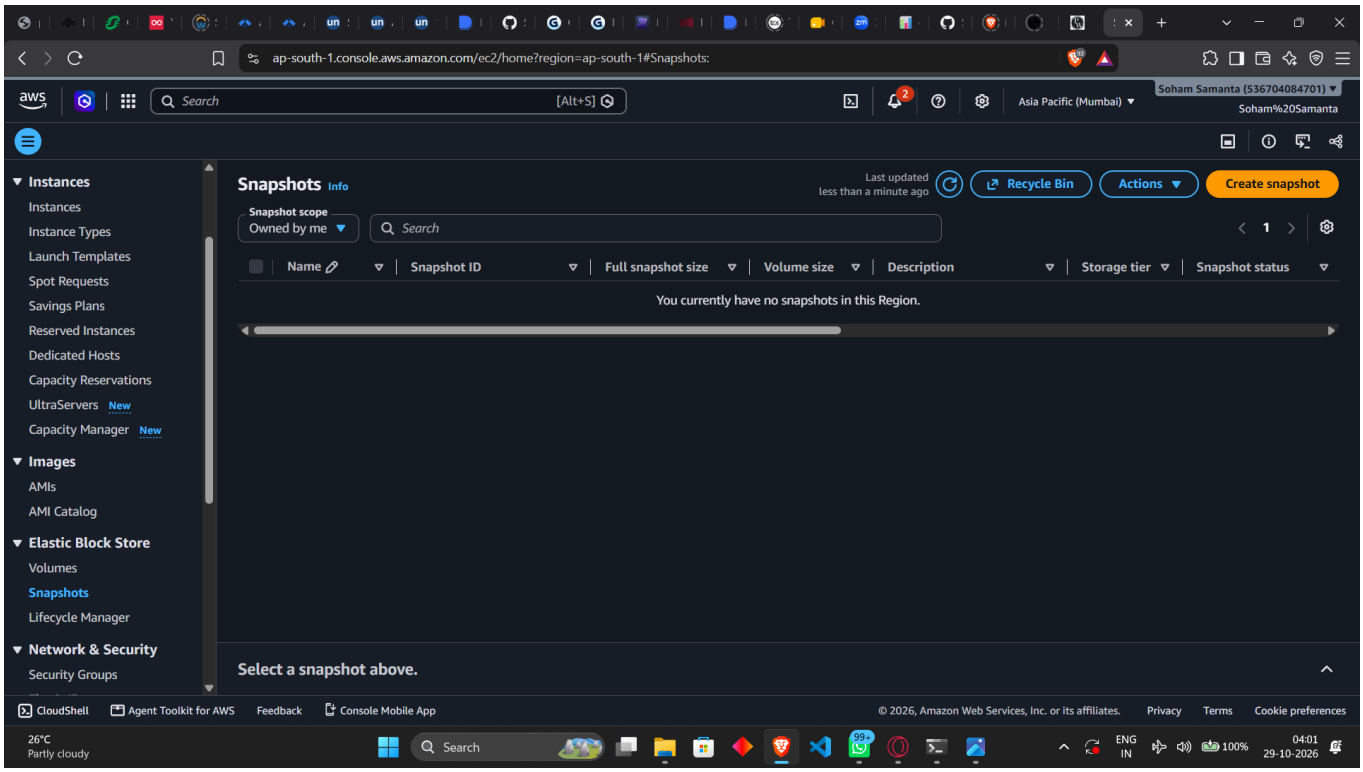


EVIDENCE PURPOSE	HOW TO READ IT
Shows no lab EBS volumes in the displayed list.	This figure supports the stage identified in the caption. It should be read with the configuration and test summaries; the report does not infer hidden values from unreadable areas.

Original screenshot: EC2 VOL CLEARED.png

Figure 28. Snapshots cleared

Shows no lab snapshots in the displayed list.

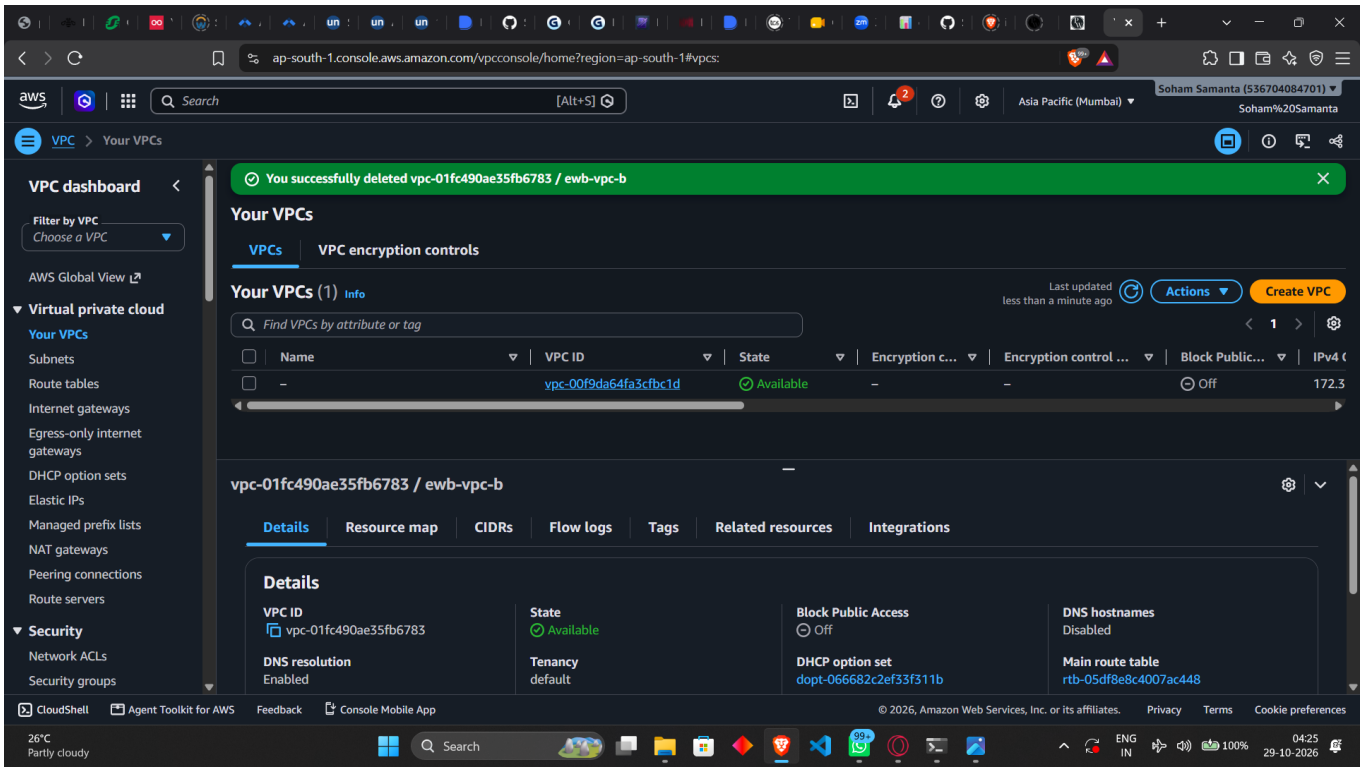


EVIDENCE PURPOSE	HOW TO READ IT
Shows no lab snapshots in the displayed list.	This figure supports the stage identified in the caption. It should be read with the configuration and test summaries; the report does not infer hidden values from unreadable areas.

Original screenshot: EC2 SNAP CLEARED.png

Figure 29. VPCs successfully removed

The supplied final VPC screenshot shows the lab VPCs are no longer present in the displayed list; the lab VPCs were successfully removed.



EVIDENCE PURPOSE	HOW TO READ IT
The supplied final VPC screenshot shows the lab VPCs are no longer present in the displayed list; the lab VPCs were successfully removed.	This figure supports the stage identified in the caption. It should be read with the configuration and test summaries; the report does not infer hidden values from unreadable areas.

Original screenshot: vpc deleted.png

9. Lessons Learned and Conclusion

VPC peering creates a private path between networks, but routing must be configured in both directions and security groups must allow the intended traffic. The successful forward and reverse pings show why return-path routing matters.

The bastion-host pattern allowed SSH administration of private instances without public IPv4 addresses on App-A and App-B. The controlled ICMP test showed that removing a required inbound rule interrupted reachability and restoring it returned connectivity.

Final checklist

- Cover details and separate architecture page
- Network inventory and route/security-group configuration
- Forward and reverse ping results
- Bastion SSH evidence
- Controlled failure and recovery evidence
- Cleanup evidence for compute, storage, networking and VPC resources

End of report